

Space–Time Algebra

A compact development of paravectors and their physical applications

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Abstract

This document develops a three-generator sigma algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$, rationalized electromagnetic units, and signature $(+, -, -, -)$ are used.

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1 The algebra at a glance

The generators σ_n , $n = 1, 2, 3$, obey ordered multiplication rules. Their products generally do not commute. The panel gives their defining relations and the general complex element developed below.

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations use

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= A_1 B_1 + A_2 B_2 + A_3 B_3, \\ \mathbf{A} \times \mathbf{B} &= \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \\ \mathbf{A}^2 &:= \mathbf{A} \cdot \mathbf{A}. \end{aligned}$$

Thus the square of a vector always denotes its self dot product. The elementary Cartesian vectors used below are the real vectors

$$\mathbf{e}_1 = \{1, 0, 0\}, \quad \mathbf{e}_2 = \{0, 1, 0\}, \quad \mathbf{e}_3 = \{0, 0, 1\}.$$

Thus $\mathbf{e}_k \cdot \boldsymbol{\sigma} = \sigma_k$; the scalar basis element is 1. Unindexed e denotes only Euler's number.

The set $\{1, \boldsymbol{\sigma}, \mathbf{i}\boldsymbol{\sigma}, \mathbf{i}\}$ is a complete basis, displayed by type as

$$\underbrace{1}_{\text{scalar}}, \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}}, \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}}, \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

Anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. The pseudoscalar is central: moving σ_n through the other two generators produces two sign changes, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}.$$

Thus its order of multiplication does not matter.

For algebra elements X, Y , the commutator is

$$\begin{aligned} [X, Y] &:= XY - YX \\ [Y, X] &= -[X, Y]. \end{aligned}$$

Thus $[X, Y] = 0$ exactly when X and Y commute. For $n, m, k \in \{1, 2, 3\}$, define the Kronecker delta by

$$\delta_{nm} := \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases}$$

and the Levi–Civita symbol by

$$\epsilon_{nmk} := \begin{cases} +1, & (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two or more indices are equal.} \end{cases}$$

Equation (1.1) gives the ordered multiplication rule for $n, m \in \{1, 2, 3\}$:

$$\begin{aligned} \sigma_n \sigma_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \\ [\sigma_n, \sigma_m] &= 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k. \end{aligned} \tag{1.1}$$

2 Hermitian and sigma-conjugate operators

Define the two operations by

$$\begin{aligned} (\cdot)^{\mathbf{H}} &:= \text{reverse the order of every product of sigma generators,} \\ (\cdot)^* &:= \text{replace every } \sigma_n \text{ by } -\sigma_n \text{ without changing product order.} \end{aligned}$$

Thus, for two generators,

$$\begin{aligned} (\sigma_m \sigma_n)^{\mathbf{H}} &= \sigma_n \sigma_m, \\ (\sigma_m \sigma_n)^* &= (-\sigma_m)(-\sigma_n). \end{aligned}$$

Both operations act term by term on sums. Their actions on the four basis types are

type representative	scalar 1	vector σ_n	bivector $\mathbf{i} \sigma_n$	pseudoscalar $\mathbf{i}$
$(\cdot)^*$	+	−	+	−
$(\cdot)^{\mathbf{H}}$	+	+	−	−
$(\cdot)^{* \mathbf{H}}$	+	−	−	+

For example, on a bivector the signs follow directly:

$$\begin{aligned} (\mathbf{i} \sigma_n)^* &= \mathbf{i}^* \sigma_n^* = \mathbf{i} \sigma_n, \\ (\mathbf{i} \sigma_n)^{\mathbf{H}} &= \sigma_n^{\mathbf{H}} \mathbf{i}^{\mathbf{H}} = -\mathbf{i} \sigma_n. \end{aligned}$$

For arbitrary algebra elements Z, Z' , the definitions give the product laws (2.1) and (2.2):

$$(ZZ')^{\mathbf{H}} = Z'^{\mathbf{H}} Z^{\mathbf{H}}, \tag{2.1}$$

$$(ZZ')^* = Z^* Z'^*. \tag{2.2}$$

The operations commute with each other: $Z^{*\mathbf{H}} = Z^{\mathbf{H}*}$. Appendix A reproduces this as Hermitian conjugation in the Pauli-matrix representation.

3 Paravectors

3.1 Core operations

Let $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. The real paravectors $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ form the general complex paravector in (3.1):

$$Z = X + iY = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.1)$$

Here $S = a + bi \in \mathbb{C}$ and $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$ are its complex scalar and vector coefficients. Equations (3.2) and (3.3) define their extraction functions:

$$\text{sc}(Z) := S, \quad (3.2)$$

$$\text{vec}(Z) := \mathbf{V}. \quad (3.3)$$

This notation applies throughout the algebra unless a real specialization is stated. On the coefficients, both conjugations give $S^* = S^H = a - bi$ and $\mathbf{V}^* = \mathbf{V}^H = \mathbf{A} - i\mathbf{B}$; on the paravector, (3.4) and (3.5) differ in the vector sign:

$$Z^* = S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.4)$$

$$Z^H = S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.5)$$

Their composition defines the adjugate in (3.6):

$$\text{adj}(Z) := Z^{*H} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.6)$$

Equation (3.7) identifies trace with twice the scalar part:

$$\text{tr}(Z) := Z + \text{adj}(Z) = 2\text{sc}(Z) = 2S. \quad (3.7)$$

The determinant in (3.8) is a complex scalar:

$$\det(Z) := \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2. \quad (3.8)$$

For complex Z' , (3.9) reverses product order. This gives determinant multiplicativity in (3.10):

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z), \quad (3.9)$$

$$\det(ZZ') = \text{adj}(Z') (\text{adj}(Z) Z) Z' = \det(Z) \det(Z'). \quad (3.10)$$

Since the determinants are scalars, the determinant of any product is unchanged by reordering its factors. For a complex scalar c , (3.11) gives the scaling law:

$$\begin{aligned} \text{adj}(cZ) &= c \text{adj}(Z), \\ \det(cZ) &= c^2 \det(Z). \end{aligned} \quad (3.11)$$

When $\det(Z) \neq 0$, equation (3.12) gives the inverse:

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \quad (3.12)$$

An invertible paravector U is called a *unit*, as in (3.13):

$$UU^{-1} = U^{-1}U = 1. \quad (3.13)$$

It is *unitary* when it satisfies (3.14):

$$U^H U = 1. \quad (3.14)$$

Then $U^H = U^{-1}$ and $U U^H = 1$. Invertibility, determinant one, and Hermitian unitarity are distinct conditions; their physical roles are compared in Section 4.

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, with complex coefficients, division is order-sensitive. An unambiguous quotient exists when Z is invertible and the factors commute, as specified in (3.15):

$$\frac{Z'}{Z} := Z' Z^{-1} = Z^{-1} Z' \quad \text{if } \mathbf{V} \times \mathbf{V}' = 0. \quad (3.15)$$

Differentiating $Z^{-1} Z = 1$ gives the inverse rule (3.16):

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (3.16)$$

Scalar part obeys the cyclic laws (3.17) and (3.18) for complex Z, Z', Z'' :

$$\text{sc}(ZZ') = \text{sc}(Z'Z), \quad (3.17)$$

$$\text{sc}(ZZ'Z'') = \text{sc}(Z'Z''Z) = \text{sc}(Z''ZZ'). \quad (3.18)$$

Multiplying these identities by 2 gives trace cyclicity. Their similarity invariants are derived in Section 4.

The real and imaginary extractions (3.19) and (3.20) use the Hermitian operation:

$$\text{re}(Z) := \frac{1}{2}(Z + Z^H) = X, \quad (3.19)$$

$$\text{im}(Z) := \frac{Z - Z^H}{2i} = Y. \quad (3.20)$$

Adjugation evaluates the extractions explicitly in (3.21) and (3.22):

$$\text{sc}(Z) = \frac{1}{2}(Z + \text{adj}(Z)) = S, \quad (3.21)$$

$$(\text{vec}(Z)) \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - \text{adj}(Z)) = \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.22)$$

The functions re and im also act on complex coefficients. Composing extractions recovers all four real parts:

$$\begin{aligned} a &= \text{re}(S) = \text{sc}(\text{re}(Z)), \\ b &= \text{im}(S) = \text{sc}(\text{im}(Z)), \\ \mathbf{A} &= \text{re}(\mathbf{V}) = \text{vec}(\text{re}(Z)), \\ \mathbf{B} &= \text{im}(\mathbf{V}) = \text{vec}(\text{im}(Z)). \end{aligned}$$

Appendix A reproduces these intrinsic operations with Pauli matrices.

3.2 Products and quadratic forms

Here $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$ are real coefficient vectors. The sigma rule (1.1) gives the dot-cross identity (3.23):

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 A_n B_m \epsilon_{nmk} i \sigma_k \\ &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (3.23)$$

Adding the product with $\mathbf{A}$ and $\mathbf{B}$ interchanged gives

$$2(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}).$$

Consequently, for every nonnegative integer n ,

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Also,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} - \mathbf{i}\mathbf{A} \times \boldsymbol{\sigma}.$$

Subtracting the two orders gives (3.24):

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.24)$$

Hence

$$\begin{aligned} [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] &= -2\mathbf{i}\mathbf{A} \times \boldsymbol{\sigma}, \\ [\mathbf{A} \cdot \mathbf{i}\boldsymbol{\sigma}, \boldsymbol{\sigma}] &= 2\mathbf{A} \times \boldsymbol{\sigma}, \\ [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} &= 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

3.3 Quadratic identity and eigenvalues

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Since $Z - S = \mathbf{V} \cdot \boldsymbol{\sigma}$ and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$,

$$\begin{aligned} (Z - S)^2 &= \mathbf{V}^2, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0. \end{aligned}$$

Using $\text{tr}(Z) = 2S$ and $\det(Z) = S^2 - \mathbf{V}^2$ gives the trace-determinant identity (3.25):

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (3.25)$$

Solving (3.25) for a scalar eigenvalue gives (3.26):

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = S \pm \sqrt{\mathbf{V}^2}. \quad (3.26)$$

Because $\mathbf{V}$ is generally complex, $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0.$$

3.4 Real products

For real $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, with $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, equation (3.23) gives the real product (3.27) and its quadratic cases:

$$\begin{aligned} XY &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ [X, Y] &= 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ X^2 &= (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (3.27)$$

Their scalar and vector parts are read directly from these equations. The real quadratic identity is (3.25) with $S = a$. If $\mathbf{A} \times \mathbf{B} = 0$, then $XY = YX$; when $\det(X) \neq 0$, also $X^{-1}Y = YX^{-1}$.

3.5 Complex products

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, where $S, S' \in \mathbb{C}$ and $\mathbf{V}, \mathbf{V}' \in \mathbb{C}^3$. Complex bilinearity extends (3.23) to the general product (3.28):

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.28)$$

Thus

$$\begin{aligned} \text{sc}(ZZ') &= SS' + \mathbf{V} \cdot \mathbf{V}', \\ \text{vec}(ZZ') &= S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}'. \end{aligned}$$

Equivalently, write $Z = X + iY$ and $Z' = X' + iY'$ with all four factors real. Their products need not be real, but distributivity gives

$$ZZ' = (XX' - YY') + i(XY' + YX').$$

Write $Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ with real $a, b, \mathbf{A}, \mathbf{B}$, and use primes for the second paravector's real components. The four real components of ZZ' are

$$\begin{aligned} \text{real scalar: } & aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \\ \text{imaginary scalar: } & ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \\ \text{real vector: } & a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \\ \text{imaginary vector: } & a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \end{aligned}$$

Specializing (3.28) gives the quadratic products:

$$\begin{aligned} Z^2 &= (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2i(ab + \mathbf{A} \cdot \mathbf{B}) \\ &\quad + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \end{aligned} \quad (3.29)$$

$$\begin{aligned} ZZ^* &= (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) - 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \\ Z^*Z &= (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) + 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \end{aligned} \quad (3.30)$$

$$ZZ^H = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.31)$$

$$Z^HZ = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.32)$$

$$ZZ^{*H} = \det(Z) = a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2i(ab - \mathbf{A} \cdot \mathbf{B}).$$

Equations (3.29) and (3.30) show $ZZ^* = Z^*Z$ exactly when $\mathbf{A} \times \mathbf{B} = 0$. Equations (3.31) and (3.32) give the same condition for $ZZ^H = Z^HZ$.

3.5.1 Norm and determinant

For the complex coefficients $S = a + bi$ and $\mathbf{V} = \mathbf{A} + i\mathbf{B}$, define

$$\begin{aligned} \|S\|^2 &:= SS^* = a^2 + b^2, \\ \|\mathbf{V}\|^2 &:= \mathbf{V} \cdot \mathbf{V}^* = \mathbf{A}^2 + \mathbf{B}^2. \end{aligned}$$

The positive paravector norm is defined by (3.33):

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2. \quad (3.33)$$

The quadratic products and cyclicity give the Hermitian form (3.34):

$$\begin{aligned}\|Z\|^2 &= \text{sc}(Z^H Z) = \frac{1}{2} \text{tr}(Z^H Z) = \text{sc}(Z Z^H) \\ &= \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*).\end{aligned}\tag{3.34}$$

For complex coefficients, S^2 and $\mathbf{V}^2$ generally differ from these positive norms. Also $\det(Z)$ may be complex. For a complex scalar $c = re^{i\theta}$, with real $r \geq 0$ and θ , scalar scaling gives

$$\begin{aligned}\det(cZ) &= r^2 e^{2i\theta} \det(Z), \\ \|cZ\|^2 &= \|c\|^2 \|Z\|^2 = r^2 \|Z\|^2.\end{aligned}\tag{3.35}$$

The determinant retains the scalar phase by (3.11); the norm loses it by (3.35).

3.6 Projection operators

For a real unit vector $\mathbf{u}$, equation (3.36) defines the two orthogonal projectors:

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).\tag{3.36}$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\begin{aligned}\Pi_{\pm}^2 &= \Pi_{\pm}, \\ \Pi_+ \Pi_- &= 0, \\ \Pi_+ + \Pi_- &= 1, \\ \Pi_{\pm}^H &= \Pi_{\pm}, \\ \Pi_{\pm}^* &= \Pi_{\mp}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} &= \pm \Pi_{\pm}.\end{aligned}$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. For a second real unit vector $\mathbf{u}'$,

$$\begin{aligned}(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ [\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. Then

$$\begin{aligned}\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \\ \Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \Pi_{\mp}.\end{aligned}$$

For $\Pi'_{\pm} = (1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})/2$, all four products are

$$\begin{aligned}\Pi_{\pm}\Pi'_{\pm} &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \pm \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \Pi_{\pm}\Pi'_{\mp} &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \mp \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

For arbitrary real vector increments $d\mathbf{u}$ and $d\mathbf{u}'$, the same dot-cross decomposition gives the geometric determinant identity

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2.\tag{3.37}$$

Dropping the nonnegative dot-product square in (3.37) gives (3.38):

$$\sqrt{\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}))} \geq \frac{1}{2} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|. \quad (3.38)$$

Equation (3.37) is Lagrange's dot-cross identity; (3.38) is a geometric bound, not a quantum uncertainty law.

3.6.1 Ladder operators

Along the third axis, equation (3.39) defines complex ladder paravectors; the real projectors specialize (3.36):

$$\begin{aligned} Z_{\pm} &= \frac{1}{2}(\sigma_1 \pm i\sigma_2) \\ \Pi_{\pm} &= \frac{1}{2}(1 \pm \sigma_3). \end{aligned} \quad (3.39)$$

Then

$$\begin{aligned} [\sigma_3, Z_{\pm}] &= \pm 2Z_{\pm}, \\ Z_{\pm}^2 &= 0, \\ Z_{\pm}Z_{\mp} &= \Pi_{\pm}, \\ \sigma_1 &= Z_+ + Z_-, \\ \sigma_2 &= (Z_+ - Z_-)/i, \\ \sigma_3 &= \Pi_+ - \Pi_-. \end{aligned}$$

The two signs give the maps between projected sectors:

$$\begin{aligned} Z_{\pm}\Pi_{\pm} &= 0, & Z_{\pm}\Pi_{\mp} &= Z_{\pm}, \\ \Pi_{\pm}Z_{\pm} &= Z_{\pm}, & \Pi_{\mp}Z_{\pm} &= 0. \end{aligned}$$

3.7 Functions of paravectors

3.7.1 Spectral rule

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$, $\mathbf{V} \in \mathbb{C}^3$, and $\mathbf{V}^2 \neq 0$. Define the complex scalars $w, \lambda_{\pm}$, complex vector $\mathbf{u}$, and paravector projectors $\Pi_{\pm}$ by

$$\begin{aligned} w &:= +\sqrt{\mathbf{V}^2}, & \lambda_{\pm} &:= S \pm w, \\ \mathbf{u} &:= \mathbf{V}/w, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

The root is the principal scalar root. The normalization is bilinear: $\mathbf{u}^2 = 1$, while the Hermitian norm $\|\mathbf{u}\|^2$ need not be 1. The identity $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ gives

$$\begin{aligned} \Pi_{\pm}^2 &= \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + 1) = \Pi_{\pm}, \\ \Pi_+\Pi_- &= \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0, \\ \Pi_+ + \Pi_- &= 1, \\ Z &= S(\Pi_+ + \Pi_-) + w(\Pi_+ - \Pi_-) = \lambda_+\Pi_+ + \lambda_-\Pi_-. \end{aligned}$$

These projectors need not be Hermitian. Orthogonality gives, for every nonnegative integer n ,

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_- = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n) \mathbf{u} \cdot \boldsymbol{\sigma}.$$

For a convergent scalar power series $f(z) = \sum_{n=0}^{\infty} c_n z^n$, with $c_n \in \mathbb{C}$, inserting these powers into $f(Z) = \sum_n c_n Z^n$ gives the spectral rule (3.40):

$$f(Z) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.40)$$

The same rule defines chosen analytic branches at the two eigenvalues. Its proof uses only complementary idempotents and therefore applies to any decomposition with these projector identities.

Adjugation exchanges the eigenvalues, so

$$\begin{aligned} \text{adj}(Z) &= \lambda_- \Pi_+ + \lambda_+ \Pi_-, \\ \det(Z) &= \lambda_+ \lambda_-, \\ Z^{-1} &= \frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-. \end{aligned}$$

The inverse requires both eigenvalues to be nonzero. Independently choosing the signs of their principal square roots gives (3.41):

$$\sqrt{Z} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-. \quad (3.41)$$

There are four distinct roots when both eigenvalues are nonzero, and two when one vanishes. Changing w to $-w$ exchanges the eigenvalue–projector pairs and leaves the function unchanged when its branch choices are exchanged with them.

3.7.2 Real paravectors and direct square roots

For real $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$, with $a \in \mathbb{R}$, $\mathbf{A} \in \mathbb{R}^3$, and $\mathbf{A} \neq 0$, the spectral quantities above reduce to

$$w = \|\mathbf{A}\|, \quad \mathbf{u} = \mathbf{A}/\|\mathbf{A}\|, \quad \lambda_{\pm} = a \pm \|\mathbf{A}\|.$$

The projectors are now Hermitian. In particular, for a real unit $\mathbf{u}$,

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1) \Pi_+ + f(-1) \Pi_- \\ &= \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1)) \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm \Pi_+ \pm i \Pi_-. \end{aligned}$$

The two root signs are independent. For a fixed analytic branch, $f(X) \rightarrow f(a)$ as $\mathbf{A} \rightarrow 0$; independent root choices require the scalar case below.

A direct component derivation also determines all *real* roots. Let $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ be real, with $b \in \mathbb{R}$, $\mathbf{B} \in \mathbb{R}^3$, and seek real $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ such that $X^2 = Y$. Equating parts gives

$$\begin{aligned} a^2 + \mathbf{A}^2 &= b, \\ 2a\mathbf{A} &= \mathbf{B}. \end{aligned}$$

If $\mathbf{B} \neq 0$, then $a \neq 0$ and $\mathbf{A} = \mathbf{B}/(2a)$. Substitution gives

$$\begin{aligned} a^4 - ba^2 + \frac{1}{4}\mathbf{B}^2 &= 0, \\ a^2 &= \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right). \end{aligned} \quad (3.42)$$

For each nonnegative a^2 in (3.42), take both scalar roots a and set $\mathbf{A} = \mathbf{B}/(2a)$; then $\mathbf{A}^2 = b - a^2$. There are four real roots when $b > \|\mathbf{B}\|$, two when $b = \|\mathbf{B}\| > 0$, and none when $b < \|\mathbf{B}\|$.

If $\mathbf{B} = 0$, the equations instead give the scalar roots $X = \pm\sqrt{b}$ and the pure-vector roots $X = \mathbf{A} \cdot \boldsymbol{\sigma}$ with $\mathbf{A}^2 = b$ for $b > 0$. For $b = 0$ only $X = 0$ is real; for $b < 0$ there is no real root.

3.7.3 Coalescing eigenvalues

For complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ with $\mathbf{V}^2 = 0$, define $N = \mathbf{V} \cdot \boldsymbol{\sigma}$, so $N^2 = 0$. A scalar function analytic at S has the terminating Taylor expansion (3.43):

$$f(Z) = f(S) + f'(S)N. \quad (3.43)$$

For $S \neq 0$, the chosen square root and logarithm are (3.44) and (3.45):

$$\sqrt{Z} = r + \frac{N}{2r}, \quad (3.44)$$

$$\log Z = \log S + \frac{N}{S}. \quad (3.45)$$

Here $r \in \mathbb{C}$ is a chosen scalar root of S ($r^2 = S$); use a consistent logarithm branch. A nonzero N with $S = 0$ has no square root: such a root would have only zero eigenvalues, and its quadratic identity would force its square to vanish. It has no logarithm because it is not invertible. If $N = 0$, a fixed analytic branch gives $f(S)$; non-scalar square roots of a scalar must instead be found from the component equations $a^2 + \mathbf{A}^2 = S$, $2a\mathbf{A} = 0$, now allowing a and $\mathbf{A}$ to be complex. Independent branch choices need not remain finite as distinct eigenvalues coalesce.

3.7.4 Exponential: projection and series derivations

For complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ with $\mathbf{V}^2 \neq 0$, recall $w = +\sqrt{\mathbf{V}^2}$, $\mathbf{u} = \mathbf{V}/w$, and $\lambda_{\pm} = S \pm w$. The spectral rule (3.40) gives

$$\begin{aligned} \exp(Z) &= \frac{1}{2}e^{S+w}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \frac{1}{2}e^{S-w}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^S \left(\frac{e^w + e^{-w}}{2} + \frac{e^w - e^{-w}}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Independently, S commutes with the vector part and $(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = w^{2n}$, $(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = w^{2n+1}\mathbf{u} \cdot \boldsymbol{\sigma}$. Separating the defining series therefore gives

$$\begin{aligned} \exp(Z) &= e^S \sum_{n=0}^{\infty} \frac{(w\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= e^S \left(\sum_{n=0}^{\infty} \frac{w^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{w^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Both derivations give (3.46):

$$\exp(Z) = e^S (\cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.46)$$

This is independent of the choice w or $-w$: $\cosh$ is even, while both $\sinh w$ and $\mathbf{u}$ change sign. When $\mathbf{V}^2 = 0$, (3.43) instead gives

$$\exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = e^S (1 + \mathbf{V} \cdot \boldsymbol{\sigma}).$$

The defining series also gives

$$\begin{aligned}\exp(Z^*) &= \exp(Z)^*, \\ \exp(Z^H) &= \exp(Z)^H.\end{aligned}$$

For the real components $a, b, \mathbf{A}, \mathbf{B}$ of $Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, write

$$w = +\sqrt{\mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}}.$$

Substituting $\mathbf{u} = (\mathbf{A} + i\mathbf{B})/w$ into (3.46) gives the component form (3.47):

$$\exp(Z) = e^a(\cos b + i \sin b) \left(\cosh w + \frac{\sinh w}{w} (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \right). \quad (3.47)$$

The real and imaginary specializations of (3.47) are

$$\exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) = e^a \left(\cosh \|\mathbf{A}\| + \frac{\sinh \|\mathbf{A}\|}{\|\mathbf{A}\|} \mathbf{A} \cdot \boldsymbol{\sigma} \right), \quad (3.48)$$

$$\exp(i(b + \mathbf{B} \cdot \boldsymbol{\sigma})) = (\cos b + i \sin b) \left(\cos \|\mathbf{B}\| + i \frac{\sin \|\mathbf{B}\|}{\|\mathbf{B}\|} \mathbf{B} \cdot \boldsymbol{\sigma} \right). \quad (3.49)$$

Setting $a = 0$ in (3.48) or $b = 0$ in (3.49) gives the pure-vector exponentials. Every quotient uses its analytic limit at a zero argument.

3.7.5 Rotation and boost factors

Let $\theta \in \mathbb{R}$ and let $\mathbf{u} \in \mathbb{R}^3$ be a real unit vector, $\mathbf{u}^2 = 1$. The two special exponentials just derived define the complex rotation factor R and the real boost factor L :

$$R(\theta, \mathbf{u}) := \exp\left(i \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.50)$$

$$L(\theta, \mathbf{u}) := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh \frac{\theta}{2} + \sinh \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.51)$$

In (3.50) and (3.51), the parameter is a rotation angle for R and a rapidity for L ; both are real dimensionless numbers. Here these names specify algebraic objects. Their actions on a general paravector are derived in Section 4; their physical applications follow in Section 9.

Because $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, reversing the parameter gives the inverses in (3.52) and (3.53):

$$R^{-1} = R(-\theta, \mathbf{u}) = \cos \frac{\theta}{2} - i \sin \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.52)$$

$$L^{-1} = L(-\theta, \mathbf{u}) = \cosh \frac{\theta}{2} - \sinh \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.53)$$

Sigma conjugation changes both i and $\boldsymbol{\sigma}$ in sign; Hermitian conjugation changes i but fixes a real vector. Hence (3.54) and (3.56) give sigma conjugates; (3.55) and (3.57) give Hermitian conjugates:

$$R^* = R, \quad (3.54)$$

$$R^H = R^{-1}, \quad (3.55)$$

$$L^* = L^{-1}, \quad (3.56)$$

$$L^H = L. \quad (3.57)$$

Equation (3.55) makes R unitary; (3.57) makes L Hermitian, so $L^H L = L^2$. Their determinants and positive norms follow from the circular and hyperbolic identities:

$$\det(R) = \cos^2(\theta/2) + \sin^2(\theta/2) = 1, \quad (3.58)$$

$$\det(L) = \cosh^2(\theta/2) - \sinh^2(\theta/2) = 1, \quad (3.59)$$

$$\|R\|^2 = 1, \quad (3.60)$$

$$\|L\|^2 = \cosh \theta. \quad (3.61)$$

Equations (3.58) and (3.59) make both adjugates equal their inverses. Equations (3.60) and (3.61) distinguish their positive norms; only R is unitary for nonzero θ . The half-angle gives the sign and period in (3.62) and (3.63):

$$R(\theta + 2\pi, \mathbf{u}) = -R(\theta, \mathbf{u}), \quad (3.62)$$

$$R(\theta + 4\pi, \mathbf{u}) = R(\theta, \mathbf{u}). \quad (3.63)$$

Gamma form. Recall $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.51). Equations (3.64) and (3.65) define its real velocity parameter and gamma factor:

$$\beta := \tanh \theta, \quad (3.64)$$

$$\gamma := \cosh \theta = \frac{1}{\sqrt{1 - \beta^2}}. \quad (3.65)$$

Here $-1 < \beta < 1$, $\gamma \geq 1$, and $\gamma\beta = \sinh \theta$. The half-angle identities give

$$\begin{aligned} \cosh \frac{\theta}{2} &= \sqrt{\frac{\gamma + 1}{2}}, \\ \sinh \frac{\theta}{2} &= \frac{\gamma\beta}{\sqrt{2(\gamma + 1)}}. \end{aligned}$$

Hence the boost factor itself is

$$L = \sqrt{\frac{\gamma + 1}{2}} \left(1 + \frac{\gamma\beta}{\gamma + 1} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.66)$$

Squaring doubles the exponent, so the full-rapidity paravector is

$$L^2 = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.67)$$

Equation (3.67) has prefactor γ ; L has the half-angle prefactor in (3.66). Equivalently, for any real vector $\mathbf{A} \neq 0$, put $\mathbf{v} = \tanh\|\mathbf{A}\| \mathbf{A}/\|\mathbf{A}\|$ and $\gamma = \cosh\|\mathbf{A}\|$. Then

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).$$

The zero-vector case follows by continuity. This last real paravector has determinant one and becomes proper velocity in Section 6.

3.7.6 Cayley transform

Let $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ be real, with $b \in \mathbb{R}$, $\mathbf{B} \in \mathbb{R}^3$, and $0 < \|\mathbf{B}\| < b$. Set $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$ and recall $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. The real conjugate $Y^* = b - \mathbf{B} \cdot \boldsymbol{\sigma} = \text{adj}(Y)$ commutes with Y , so

$$\frac{Y}{Y^*} = \frac{Y^2}{\det(Y)} = \frac{b^2 + \mathbf{B}^2 + 2b\mathbf{B} \cdot \boldsymbol{\sigma}}{b^2 - \mathbf{B}^2}.$$

Define the real parameter $\theta = 2 \tanh^{-1}(\|\mathbf{B}\|/b)$. The double-angle identities give

$$\begin{aligned} \cosh \theta &= \frac{b^2 + \mathbf{B}^2}{b^2 - \mathbf{B}^2}, \\ \sinh \theta &= \frac{2b\|\mathbf{B}\|}{b^2 - \mathbf{B}^2}. \end{aligned}$$

Thus the quotient is the full-rapidity exponential

$$\begin{aligned} \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \Pi_+ + e^{-\theta} \Pi_-. \end{aligned} \quad (3.68)$$

Equation (3.68) is $L(\theta, \mathbf{u})^2$ from (3.67). Its real generator vector is $\mathbf{A} = \theta \mathbf{u}$, while the Cayley vector obeys

$$\mathbf{B} = b \tanh\left(\frac{\|\mathbf{A}\|}{2}\right) \frac{\mathbf{A}}{\|\mathbf{A}\|}. \quad (3.69)$$

Equation (3.69) gives $\|\mathbf{B}\|/b = \tanh(\theta/2)$, the half-rapidity parameter. The corresponding real gamma factor and vector are

$$\begin{aligned} \gamma &= \frac{b^2 + \mathbf{B}^2}{b^2 - \mathbf{B}^2}, \\ \mathbf{v} &= \frac{2b\mathbf{B}}{b^2 - \mathbf{B}^2}. \end{aligned}$$

They give the equivalent forms

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} = \frac{Y}{Y^*}.$$

Choosing $b = 1$ gives the normalized Cayley form; $\mathbf{B} = 0$ follows by continuity.

3.7.7 Composing the factors

Recall $L(\theta, \mathbf{u}) = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $R(\theta, \mathbf{u}) = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.51) and (3.50). For a fixed real unit direction $\mathbf{u}$ and real parameters θ, θ' , the generators commute. Equation (3.70) therefore adds their parameters:

$$L(\theta, \mathbf{u})L(\theta', \mathbf{u}) = L(\theta + \theta', \mathbf{u}). \quad (3.70)$$

$$\begin{aligned} R(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta + \theta', \mathbf{u}), \\ L(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta', \mathbf{u})L(\theta, \mathbf{u}). \end{aligned}$$

For the aligned real and imaginary generator, put $w = (\theta + i\theta')/2 \in \mathbb{C}$. Its exponential is

$$\exp(w\mathbf{u} \cdot \boldsymbol{\sigma}) = L(\theta, \mathbf{u})R(\theta', \mathbf{u}) = \cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}.$$

Expanding the complex scalar coefficients gives

$$\begin{aligned}\cosh w &= \cosh \frac{\theta}{2} \cos \frac{\theta'}{2} + i \sinh \frac{\theta}{2} \sin \frac{\theta'}{2}, \\ \sinh w &= \sinh \frac{\theta}{2} \cos \frac{\theta'}{2} + i \cosh \frac{\theta}{2} \sin \frac{\theta'}{2}.\end{aligned}$$

This aligned factorization requires parallel real and imaginary generator vectors. For different real unit directions $\mathbf{u}, \mathbf{v}$, the vector product instead gives

$$\left[\frac{1}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, \frac{1}{2}\mathbf{v} \cdot \boldsymbol{\sigma}\right] = \frac{i}{2}(\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (3.71)$$

Equation (3.71) is a rotation generator: boosts about different axes do not commute.

3.7.8 Logarithm

For complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ with $\mathbf{V}^2 \neq 0$, recall $w = +\sqrt{\mathbf{V}^2}$, $\mathbf{u} = \mathbf{V}/w$, $\lambda_{\pm} = S \pm w$, and $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Assume both eigenvalues are nonzero. For a nonzero complex scalar λ , the scalar branches are

$$\log \lambda = \log \|\lambda\| + i(\arg \lambda + 2n\pi), \quad n \in \mathbb{Z}.$$

Choose one branch for each eigenvalue. The spectral rule gives (3.72):

$$\begin{aligned}\log Z &= \log(\lambda_+)\Pi_+ + \log(\lambda_-)\Pi_- \\ &= \frac{1}{2}(\log \lambda_+ + \log \lambda_-) + \frac{1}{2}(\log \lambda_+ - \log \lambda_-)\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned} \quad (3.72)$$

Exchanging w and $-w$ exchanges both eigenvalue–projector pairs and preserves the same chosen logarithm. Independent scalar branches give the different logarithms.

For a real unit vector $\mathbf{u}$, (3.72) uses eigenvalues 1 and -1 , giving (3.73) for integers n, m :

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i \Pi_+ + (2m+1)\pi i \Pi_-. \quad (3.73)$$

For real $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, with $b \in \mathbb{R}$, $\mathbf{B} \in \mathbb{R}^3$, and $b > \|\mathbf{B}\|$, both eigenvalues are positive. Combining their real logarithms yields (3.74):

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \tanh^{-1}\left(\frac{\|\mathbf{B}\|}{b}\right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.74)$$

The vector quotient uses its limit at $\mathbf{B} = 0$; the general coalescing case was treated above.

4 Transformations and invariants

Here $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ is a complex paravector: $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Transformation factors are constant complex paravectors. Distinguish the scalar part $\text{sc}(Z) = S = \frac{1}{2} \text{tr} Z$ from (3.7), the complex determinant $S^2 - \mathbf{V}^2$ from (3.8), and the positive norm $\text{sc}(Z^H Z)$ from (3.34). Scalar part and trace carry exactly the same information; we use sc below.

4.1 Similarity, determinant, and norm

For an invertible complex paravector T , similarity means $Z' = T^{-1}ZT$. Scalar-part cyclicity proves (4.1):

$$\text{sc}(Z') = \text{sc}(T^{-1}ZT) = \text{sc}(ZTT^{-1}) = \text{sc}(Z). \quad (4.1)$$

Thus trace is also preserved. Multiplicativity proves the determinant invariance (4.2):

$$\det(Z') = \det(T^{-1}) \det(Z) \det(T) = \det(Z). \quad (4.2)$$

The opposite similarity $Z' = TZZ^{-1}$ has the same invariants. For independent complex factors T, T' , the law $Z' = T'ZT$ instead gives

$$\det(Z') = \det(T') \det(T) \det(Z). \quad (4.3)$$

Equation (4.3) preserves the determinant for every Z exactly when $\det(T') \det(T) = 1$. This alone does not preserve scalar part, norm, or the real paravector subspace.

Recall that a unitary U satisfies $U^H = U^{-1}$ by (3.14). For $Z' = U^H Z U$, cancellation and scalar-part cyclicity prove (4.4):

$$\begin{aligned} \|Z'\|^2 &= \text{sc}(U^H Z^H U U^H Z U) \\ &= \text{sc}(Z^H Z U U^H) = \|Z\|^2. \end{aligned} \quad (4.4)$$

The norm is also preserved by $Z' = UZ$, $Z' = ZU$, and $Z' = UZU^H$. The two similarities additionally preserve scalar part and determinant; a one-sided unitary product need not.

4.1.1 Unit determinant is not Hermitian unitarity

The word *unit* in (3.13) means invertible; *unitary* means (3.14). A useful third condition is determinant one. To see their difference, let $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$ be a real paravector increment, with $dt > \|d\mathbf{r}\|$, and define the positive real scalar $ds = \sqrt{dt^2 - d\mathbf{r}^2}$. The real normalized paravector is $U = dX/ds$. Since $dX^* dX = ds^2$,

$$U^* U = \frac{dX^* dX}{ds^2} = 1. \quad (4.5)$$

Equation (4.5) gives $\det(U) = 1$ and $U^{-1} = U^*$. However, the two conjugates are

$$\begin{aligned} U^H &= U = \frac{dt + d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds}, \\ U^* &= \frac{dt - d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds}. \end{aligned}$$

Hermitian conjugation leaves every real paravector fixed; sigma conjugation reverses its vector part. Therefore

$$U^H U = U^2 = \frac{dt^2 + d\mathbf{r}^2 + 2dt \, d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds^2}, \quad (4.6)$$

$$\|U\|^2 = \frac{dt^2 + d\mathbf{r}^2}{dt^2 - d\mathbf{r}^2}. \quad (4.7)$$

Both expressions in (4.6) and (4.7) equal 1 only when $d\mathbf{r} = 0$. In Section 6, this U is proper velocity: its normalization is $U^* U = 1$, while $U^H = U^*$ holds only in its rest frame.

4.2 Real paravectors and the Lorentz metric

Let $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ be real, with $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$. Equation (3.8) gives $\det(X) = a^2 - \mathbf{A}^2$, with Lorentz signature; (3.33) gives $\|X\|^2 = a^2 + \mathbf{A}^2$, with Euclidean signature. A transformation of real spacetime paravectors must preserve both reality and the determinant. For a complex T with $\det(T) = 1$, these requirements hold for the Hermitian congruence

$$\boxed{X' = T^{-1} X (T^{-1})^H.} \quad (4.8)$$

Equation (4.8) preserves reality and determinant because

$$\begin{aligned} X'^H &= T^{-1} X^H (T^{-1})^H = X', \\ \det(X') &= \det(T^{-1}) \det(X) \det((T^{-1})^H) = \det(X). \end{aligned}$$

These congruences give the proper, future-preserving Lorentz transformations. In the matrix representation, the eigenvalues of X are $a \pm \|\mathbf{A}\|$; the future cone consists of positive semidefinite Hermitian matrices. Congruence preserves this positivity. The factors T and $-T$ give the same transformation, producing the double cover $\text{SL}(2, \mathbb{C})$.^[1]

For real θ and a real unit $\mathbf{u}$, the algebraic definitions $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ reduce this congruence to $X' = R^H X R$ and $X' = L^{-1} X L^{-1}$, respectively. Their components are derived in the next two subsections. For $X = 1$, equation (4.9) makes the invariant distinction explicit:

$$\begin{aligned} X' &= L^{-2} = \cosh \theta - \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \det(X') &= 1, \\ \text{sc}(X') &= \cosh \theta, \\ \|X'\|^2 &= \cosh(2\theta). \end{aligned} \quad (4.9)$$

Thus the determinant supplies the spacetime metric, while the object being transformed determines whether to use congruence or similarity.

4.3 Rotation action and Rodrigues' formula

Recall the unitary factor $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.50), with real angle θ and real unit axis $\mathbf{u}$. For the complex paravector $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$Z' = R^H Z R = R^{-1} Z R. \quad (4.10)$$

Equation (4.10) keeps $S' = S$. The vector product (3.23) gives

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2(\mathbf{u} \cdot \mathbf{V})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{V} \cdot \boldsymbol{\sigma}, \\ [\mathbf{V} \cdot \boldsymbol{\sigma}, \mathbf{u} \cdot \boldsymbol{\sigma}] &= -2i(\mathbf{u} \times \mathbf{V}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Inserting the sine and cosine forms of R and R^{-1} therefore yields

$$\begin{aligned} \mathbf{V}' \cdot \boldsymbol{\sigma} &= \cos^2 \frac{\theta}{2} \mathbf{V} \cdot \boldsymbol{\sigma} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} (\mathbf{u} \times \mathbf{V}) \cdot \boldsymbol{\sigma} \\ &\quad + \sin^2 \frac{\theta}{2} (2(\mathbf{u} \cdot \mathbf{V})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Collecting terms gives Rodrigues' formula:

$$\boxed{\mathbf{V}' = \mathbf{V} \cos \theta + (\mathbf{u} \times \mathbf{V}) \sin \theta + (\mathbf{u} \cdot \mathbf{V})(1 - \cos \theta)\mathbf{u}.} \quad (4.11)$$

Equation (4.11) is a right-handed rotation through $+\theta$ about $\mathbf{u}$. It preserves sc, determinant, norm, and the real paravector subspace. For complex $\mathbf{V}$, it rotates its real and imaginary parts by the same real rotation. The sign change in (3.62) affects both factors in (4.10), so Z' has period 2π .

4.4 Boost action and its sign convention

Recall $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.51); $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, $\beta = \tanh \theta$, and $\gamma = \cosh \theta$. For a complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, choose the congruence

$$Z' = L^{-1} Z L^{-1}. \quad (4.12)$$

The factors in (4.12) have the same sign because $L^H = L$ by (3.57). Write the complex vector as the sum of its parallel and perpendicular parts:

$$\begin{aligned} \mathbf{V}_{\parallel} &:= (\mathbf{u} \cdot \mathbf{V})\mathbf{u}, \\ \mathbf{V}_{\perp} &:= \mathbf{V} - \mathbf{V}_{\parallel}. \end{aligned}$$

The parallel term commutes with $\mathbf{u} \cdot \boldsymbol{\sigma}$; the perpendicular term anticommutes with it. Consequently,

$$\begin{aligned} L^{-1}(S + \mathbf{V}_{\parallel} \cdot \boldsymbol{\sigma})L^{-1} &= (S + \mathbf{V}_{\parallel} \cdot \boldsymbol{\sigma})L^{-2}, \\ L^{-1}(\mathbf{V}_{\perp} \cdot \boldsymbol{\sigma})L^{-1} &= \mathbf{V}_{\perp} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Using $L^{-2} = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma})$ from (3.67), multiplication gives

$$\boxed{S' = \gamma(S - \beta \mathbf{u} \cdot \mathbf{V}).} \quad (4.13)$$

$$\boxed{\mathbf{V}' = \mathbf{V}_{\perp} + \gamma(\mathbf{V}_{\parallel} - \beta S \mathbf{u}).} \quad (4.14)$$

Equations (4.13) and (4.14) preserve $S^2 - \mathbf{V}^2$ and reality; they generally change S and $\|\mathbf{V}\|^2$.

The inverse signs are a convention: equation (3.53) gives $L^{-1} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = L(-\theta, \mathbf{u})$. Replacing θ by $-\theta$ changes (4.12) to $Z' = L Z L$ and changes β to $-\beta$ in the component equations. The two conventions describe opposite boosts. We keep the positive exponential definition of L throughout; in the spacetime application, (4.12) gives components in the frame moving with velocity $+\beta \mathbf{u}$.

A boost similarity is a different action. For $Z' = L^{-1} Z L$, the same commutation rules instead give $S' = S$ and

$$\mathbf{V}' = \mathbf{V}_{\parallel} + \gamma(\mathbf{V}_{\perp} - i\beta \mathbf{u} \times \mathbf{V}). \quad (4.15)$$

The similarity (4.15) preserves scalar part and determinant, but a real vector generally acquires an imaginary component. It will act on the complex electromagnetic field, whose real and imaginary vectors represent two physical fields.

5 Differentiated fields

5.1 Paravector differential operators

Let $t \in \mathbb{R}$ and $\mathbf{r} = \{r_1, r_2, r_3\} \in \mathbb{R}^3$ be real coordinates. Define their scalar and vector derivatives by

$$\partial_t := \frac{\partial}{\partial t}, \quad \partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}.$$

Equations (5.1) and (5.2) define the partial paravector and its sigma conjugate:

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sum_{n=1}^3 \sigma_n \frac{\partial}{\partial r_n}, \quad (5.1)$$

$$\partial^* = \text{adj}(\partial) = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}. \quad (5.2)$$

The sigma generators are constant. Derivatives act from the left on everything to their right unless parentheses specify their scope. For fields with commuting second derivatives, anticommutation cancels all mixed spatial terms:

$$\begin{aligned} (\boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})^2 &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} + \sum_{n < m} (\sigma_n \sigma_m + \sigma_m \sigma_n) \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

The mixed time-space terms likewise cancel. Both orders therefore give the scalar d'Alembertian in (5.3):

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}} =: \square. \quad (5.3)$$

These are operator identities: $\partial \partial^* Z$ means $\partial(\partial^* Z)$.

5.2 Components and the product rule

Let the complex paravector field be

$$Z(t, \mathbf{r}) = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma},$$

with real scalar functions $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$ and real vector functions $\mathbf{A}, \mathbf{B} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. Ordered vector multiplication gives its four real components in (5.4):

$$\begin{aligned} \partial Z &= (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \\ &\quad + (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\partial_{\mathbf{r}} b + \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (5.4)$$

Multiplication by i exchanges real and imaginary parts, with the former imaginary parts acquiring a minus sign.

For a second complex paravector field $Z'(t, \mathbf{r})$, the prime labels the field rather than a derivative. Apply the scalar product rule before reordering any factors:

$$\begin{aligned}\partial(ZZ') &= (\partial_t Z)Z' + Z(\partial_t Z') \\ &\quad + \sum_{n=1}^3 \sigma_n \left(\frac{\partial Z}{\partial r_n} Z' + Z \frac{\partial Z'}{\partial r_n} \right) \\ &= (\partial Z)Z' + Z(\partial Z') + \sum_{n=1}^3 [\sigma_n, Z] \frac{\partial Z'}{\partial r_n}.\end{aligned}$$

Define the last sum as $[\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z' := \sum_n [\sigma_n, Z] \partial Z' / \partial r_n$. Changing the spatial sign gives the paired rules (5.5) and (5.6):

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z', \quad (5.5)$$

$$\partial^*(ZZ') = (\partial^* Z)Z' + Z(\partial^* Z') - [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (5.6)$$

The contraction retains the displayed multiplied order.

For a noncommuting check, take $Z = \sigma_2$ and $Z' = r_1$. Since $[\sigma_1, \sigma_2] = 2i\sigma_3$, the two right-hand sides give

$$\begin{aligned}\partial(\sigma_2 r_1) &= \sigma_2 \sigma_1 + [\sigma_1, \sigma_2] = -i\sigma_3 + 2i\sigma_3 = i\sigma_3, \\ \partial^*(\sigma_2 r_1) &= -\sigma_2 \sigma_1 - [\sigma_1, \sigma_2] = i\sigma_3 - 2i\sigma_3 = -i\sigma_3.\end{aligned}$$

These agree with direct differentiation, $\pm\sigma_1\sigma_2$. The ordinary Leibniz rule holds when the left factor commutes with every σ_n , in particular for a complex scalar. Otherwise the commutator term must be retained.

6 Space–time kinematics

Let $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ be a real event, with time $t \in \mathbb{R}$ and position $\mathbf{r} \in \mathbb{R}^3$. A massive particle follows a future-directed time-like path $\mathbf{r}(t)$; s denotes its proper time.

6.1 Intervals, velocity, and momentum

The differential event and its invariant interval are

$$\begin{aligned}dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}. \\ ds^2 &= \det(dX) = dX \, dX^* = dt^2 - d\mathbf{r}^2.\end{aligned} \quad (6.1)$$

By (4.8), a constant complex factor T with $\det T = 1$ gives

$$dX' = T^{-1} dX (T^{-1})^H, \quad \det(dX') = ds^2.$$

Positive, zero, and negative values of (6.1) mean time-like, null, and space-like separation. Massive particles follow time-like paths; light follows null paths; causal signals cannot connect space-like separated events. For two real intervals,

$$\begin{aligned}[dX, dX'] &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}, \\ (dX)^{-1} &= \frac{dX^*}{ds^2} \quad (ds^2 \neq 0).\end{aligned}$$

A null interval has no inverse.

Along the massive path, define the real coordinate velocity $\mathbf{v} = d\mathbf{r}/dt$ and choose $ds > 0$. Then $ds^2 = (1 - \mathbf{v}^2) dt^2$ gives

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{dt}{ds}, \quad (6.2)$$

$$\frac{d}{ds} = \gamma \frac{d}{dt}. \quad (6.3)$$

Applying (6.3) with (6.2) gives the real proper velocity

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.4)$$

For (6.4), the normalization (4.5) gives $U^*U = 1$ and $U^{-1} = U^* = \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma})$. Since $U^H = U$, this is unit Minkowski determinant, not Hermitian unitarity except at rest. For a real unit direction $\mathbf{u}$, write $\mathbf{v} = \beta\mathbf{u}$ and $\theta = \tanh^{-1} \beta$. Equation (3.67) gives $U = L^2$ for $L = \exp(\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$. Between real times t_0 and t , the elapsed proper time is

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')}. \quad (6.5)$$

Equation (6.5) compares clock readings along the path. For constant real mass $m > 0$, define the real four-momentum

$$P := mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}. \quad (6.6)$$

The real coefficients of (6.6) are energy and momentum:

$$\begin{aligned} E &= \text{sc}(P) = m\gamma, \\ \mathbf{p} &= \text{vec}(P) = m\gamma\mathbf{v}, \\ P^{-1} &= \frac{E - \mathbf{p} \cdot \boldsymbol{\sigma}}{m^2}. \end{aligned}$$

The determinant gives the mass shell

$$E^2 - \mathbf{p}^2 = \det P = m^2. \quad (6.7)$$

Equation (6.7) follows from $\det U = 1$. A massless path has no proper-time velocity. Define its momentum directly as a future null paravector $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, with $E > 0$ and $E^2 = \mathbf{p}^2$; it has no inverse.

6.2 Proper acceleration

Recall the real velocity $\mathbf{v}(t)$, proper velocity U , and Lorentz factor γ above. Define the real coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$. Differentiating gives

$$\begin{aligned} \frac{d\gamma}{dt} &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2}(-2\mathbf{v} \cdot \mathbf{a}) = \gamma^3 \mathbf{v} \cdot \mathbf{a}, \\ \frac{d^2}{ds^2} &= \gamma \frac{d}{dt} \left(\gamma \frac{d}{dt} \right) = \gamma^2 \frac{d^2}{dt^2} + \gamma^4 (\mathbf{v} \cdot \mathbf{a}) \frac{d}{dt}. \end{aligned}$$

Applying this to X yields the real proper-acceleration paravector:

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^2 \mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.8)$$

Differentiating $U^*U = 1$ gives

$$\left(\frac{dU}{ds}\right)^* U + U^* \frac{dU}{ds} = 0.$$

The two terms have equal real scalar parts, so

$$\text{sc}(U^* dU/ds) = 0. \quad (6.9)$$

Equation (6.9) states Minkowski orthogonality of four-velocity and four-acceleration. Multiplication on the right by U^* also gives

$$\begin{aligned} \text{adj}\left(\frac{dU}{ds}\right) &= -U^* \frac{dU}{ds} U^*. \\ U^* \frac{dU}{ds} &= \gamma^3(\mathbf{a} - i\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (6.10)$$

In (6.10), the real vector records acceleration and the imaginary vector records turning of the velocity. Since $\mathbf{a} \cdot (\mathbf{v} \times \mathbf{a}) = 0$,

$$\begin{aligned} \det\left(\frac{dU}{ds}\right) &= -\left(U^* \frac{dU}{ds}\right)^2 \\ &= -\gamma^6(\mathbf{a}^2 - (\mathbf{v} \times \mathbf{a})^2) \\ &= -\gamma^4(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2). \end{aligned} \quad (6.11)$$

The last equality in (6.11) uses $(\mathbf{v} \times \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}^2 - (\mathbf{v} \cdot \mathbf{a})^2$ and $1 - \mathbf{v}^2 = \gamma^{-2}$. Nonzero proper acceleration is therefore space-like; its invariant squared magnitude is the negative determinant. For $\mathbf{v} \neq 0$, define $\mathbf{a}_{\parallel} = (\mathbf{v} \cdot \mathbf{a})\mathbf{v}/\mathbf{v}^2$ and $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}$. Equivalently,

$$-\det\left(\frac{dU}{ds}\right) = \gamma^4 \mathbf{a}_{\perp}^2 + \gamma^6 \mathbf{a}_{\parallel}^2. \quad (6.12)$$

At rest (6.12) reduces to $\mathbf{a}^2$.

6.3 Free-particle action

Recall real $m > 0$, $\mathbf{v} = d\mathbf{r}/dt$, and $ds = dt/\gamma$. Here $L(t, \mathbf{r}, \mathbf{v})$ is a real scalar Lagrangian, not the boost factor. The free action is the physical choice

$$\int L dt = -m \int ds. \quad (6.13)$$

Using $ds = dt/\gamma$, equation (6.13) gives the Lagrangian

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m/\gamma. \quad (6.14)$$

For endpoint-fixed real path variations $\delta\mathbf{r}(t)$, varying (6.14) and integrating by parts gives

$$\delta \int L dt = \int \left(\partial_{\mathbf{r}} L - \frac{d}{dt} \partial_{\mathbf{v}} L \right) \cdot \delta\mathbf{r} dt.$$

Stationarity for every variation yields the Euler–Lagrange equation. With real Hamiltonian $H := \mathbf{v} \cdot \partial_{\mathbf{v}} L - L$, the chain rule gives

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L, \quad (6.15)$$

$$\frac{dH}{dt} = \mathbf{v} \cdot \left(\frac{d}{dt} \partial_{\mathbf{v}} L - \partial_{\mathbf{r}} L \right) - \partial_t L = -\partial_t L. \quad (6.16)$$

By (6.15) and (6.16), spatial and temporal translation invariance conserve canonical momentum and energy. For the free action,

$$\begin{aligned} \partial_{\mathbf{v}} L &= m\gamma \mathbf{v} = \mathbf{p}, \\ H &= m\gamma \mathbf{v}^2 + m/\gamma = m\gamma = E. \end{aligned}$$

Writing $v = \|\mathbf{v}\| \ll 1$, the binomial series gives

$$\begin{aligned} \sqrt{1 - v^2} &= 1 - \frac{1}{2}v^2 + O(v^4), \\ \gamma &= 1 + \frac{1}{2}v^2 + O(v^4), \\ L &= -m + \frac{1}{2}mv^2 + O(mv^4), \\ H &= m + \frac{1}{2}mv^2 + O(mv^4). \end{aligned}$$

The exact kinetic energy above rest is

$$K := H - m = m(\gamma - 1). \quad (6.17)$$

Equation (6.17) and both expansions recover $mv^2/2$ at low speed after removal of the constant rest term.

7 Electromagnetism

All coefficient fields in this section are real functions of real $(t, \mathbf{r})$. Define

$$\begin{aligned} F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} && \text{electric and magnetic fields,} \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma} && \text{scalar and vector potentials,} \\ D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} && \text{charge and current densities.} \end{aligned} \quad (7.1)$$

Thus Φ, D are real paravectors, while (7.1) is a complex pure vector. Recall $\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}$ and $\partial^* \partial = \square = \partial_t^2 - \partial_{\mathbf{r}}^2$.

The dot/cross product (3.23) combines metric and orientation. With a spatial derivative it gives divergence and curl, while the six real components of F represent an antisymmetric spacetime field tensor. This algebra of physical space is isomorphic to the even subalgebra of spacetime algebra; its real paravectors represent spacetime vectors relative to an observer.[1] The product explains the compact form; Maxwell dynamics remains physical input, derived from an action in Section 7.3.

7.1 Maxwell's equations and wave propagation

Maxwell's four equations are the single field law

$$\boxed{\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.} \quad (7.2)$$

The product rule expands the left side of (7.2) into

$$\begin{aligned}\partial F = & \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{\mathrm{i} \partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \\ & + \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \\ & + \underbrace{\mathrm{i}(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}.\end{aligned}$$

Equating the four parts gives

$$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (7.3)$$

$$\partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (7.4)$$

$$\partial_{\mathbf{r}} \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}, \quad (7.5)$$

$$\partial_{\mathbf{r}} \times \mathbf{E} + \partial_t \mathbf{B} = 0. \quad (7.6)$$

Equations (7.3)–(7.6) are Gauss’ electric and magnetic laws, Ampère–Maxwell’s law, and Faraday’s law. Applying ∂^* to (7.2) gives

$$\begin{aligned}\square F &= \partial^* D^* = (\partial D)^* \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + \mathrm{i} \partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}.\end{aligned} \quad (7.7)$$

The scalar part of (7.7) vanishes because F is scalar-free. Thus

$$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (7.8)$$

$$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (7.9)$$

$$\square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (7.10)$$

Equation (7.8) is charge conservation; (7.9) and (7.10) are the sourced field waves. In vacuum, $D = 0$ and both fields obey the free wave equation.

7.2 Potentials and gauge freedom

Recall the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. Its derivative is

$$\partial \Phi = (\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}) + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma} + \mathrm{i}(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.$$

Define the real scalar $S := \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. Identifying the remaining components with F^* gives

$$\partial \Phi = S + F^*, \quad (7.11)$$

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \quad (7.12)$$

$$\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (7.13)$$

Equation (7.11) separates the gauge scalar from the field; (7.12) and (7.13) satisfy (7.4) and (7.6) identically. Since $F = \partial^* \Phi^* - S$,

$$\begin{aligned}\partial F &= \square \Phi^* - \partial S \\ &= (\square V - \partial_t S) - (\square \mathbf{A} + \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Comparison with (7.2) yields

$$\rho = \square V - \partial_t S, \quad (7.14)$$

$$\mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (7.15)$$

These source equations (7.14)–(7.15) hold before gauge fixing. For a real scalar gauge function $\lambda(t, \mathbf{r})$, let

$$\Phi' = \Phi + \partial^* \lambda. \quad (7.16)$$

Differentiating (7.16) gives $\partial \Phi' = \partial \Phi + \square \lambda$. Only the scalar changes: $S' = S + \square \lambda$, while $F' = F$. In the Lorenz gauge $S = 0$, the source equations reduce to $\square V = \rho$ and $\square \mathbf{A} = \mathbf{J}$.

7.3 The electromagnetic field action

For real scalar potential $V(t, \mathbf{r})$ and vector potential $\mathbf{A}(t, \mathbf{r})$, write $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. With prescribed D , Maxwell dynamics follows from the real Lagrangian density

$$L = \frac{1}{2} \text{re}(F^2) - \text{sc}(\Phi^* D) = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) - \rho V + \mathbf{J} \cdot \mathbf{A}. \quad (7.17)$$

In varying the action of (7.17), boundary-vanishing variations satisfy

$$\begin{aligned} \delta \mathbf{E} &= -\partial_{\mathbf{r}} \delta V - \partial_t \delta \mathbf{A}, \\ \delta \mathbf{B} &= \partial_{\mathbf{r}} \times \delta \mathbf{A}. \end{aligned}$$

Integration by parts gives

$$\delta \int L \, dt \, d^3 \mathbf{r} = \int [(\partial_{\mathbf{r}} \cdot \mathbf{E} - \rho) \delta V + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} + \mathbf{J}) \cdot \delta \mathbf{A}] \, dt \, d^3 \mathbf{r}. \quad (7.18)$$

Independent variations in (7.18) give (7.3) and (7.5); the potential identities supply (7.4) and (7.6). Together they recover (7.2). The symbol δ denotes variation.

7.4 Field invariants

Since $\text{sc}(F) = 0$, its adjugate is $-F$. Squaring gives

$$F^2 = -\det F = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}. \quad (7.19)$$

For an invertible complex paravector T , the field similarity $F' = T^{-1}FT$ preserves this scalar by (4.2). Physical rotations and boosts use $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, respectively, with real unit $\mathbf{u}$ and real angle or rapidity θ ; their field laws are derived in Section 9. The real and imaginary parts of (7.19) are therefore observer-independent. For a nonzero field, they give the classification

Condition	Consequence
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 > \mathbf{B}^2$	Some frame has $\mathbf{B}' = 0$.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 < \mathbf{B}^2$	Some frame has $\mathbf{E}' = 0$.
$\mathbf{E} \cdot \mathbf{B} \neq 0$	Neither field can be eliminated.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 = \mathbf{B}^2$	Null field: equal, perpendicular fields.

7.5 Lorentz force from the action

Let $q \in \mathbb{R}$ be charge and $m > 0$ constant mass. Recall the real worldline quantities $U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ and $P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, with $E = m\gamma$ and $\mathbf{p} = m\gamma\mathbf{v}$. The real particle Lagrangian L is fixed by the action

$$\int L dt = -m \int ds - q \int \text{sc}(\Phi^* U) ds. \quad (7.20)$$

Since $\text{sc}(\Phi^* U) = \gamma(V - \mathbf{v} \cdot \mathbf{A})$, (7.20) gives

$$L = -m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}. \quad (7.21)$$

The Euler–Lagrange terms of (7.21) are

$$\begin{aligned} \partial_v L &= \mathbf{p} + q\mathbf{A}, \\ \frac{d}{dt} \partial_v L &= \frac{d\mathbf{p}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_r) \mathbf{A}, \\ \partial_r L &= -q\partial_r V + q\partial_r(\mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

Here spatial differentiation holds the independent velocity argument fixed. Using $\partial_r(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_r) \mathbf{A} = \mathbf{v} \times (\partial_r \times \mathbf{A})$ gives

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.22)$$

Here (7.12) and (7.13) turn the potential derivatives into the fields of (7.22). The Hamiltonian and its balance law are

$$H = \mathbf{v} \cdot (\mathbf{p} + q\mathbf{A}) - L = E + qV. \quad (7.23)$$

Using (6.16) in (7.23),

$$\begin{aligned} \frac{dE}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_r V &= -\partial_t L = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \\ \frac{dE}{dt} &= q\mathbf{v} \cdot \mathbf{E}. \end{aligned} \quad (7.24)$$

Equation (7.24) is the mechanical work rate. Thus $\mathbf{p} + q\mathbf{A}$ and H are canonical momentum and energy; $\mathbf{p}, E$ are mechanical momentum and energy, with E including rest energy. Canonical energy is conserved for static potentials in the chosen gauge. A time-dependent gauge transformation can change H while leaving E and the force unchanged.

Now expand the complex product of the field and proper velocity:

$$\begin{aligned} \frac{FU}{\gamma} &= \mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} - \mathbf{v} \times \mathbf{E}) \cdot \boldsymbol{\sigma}). \end{aligned} \quad (7.25)$$

The real scalar and vector parts of (7.25) reproduce (7.24) and (7.22). Since P is real and $d/ds = \gamma d/dt$, they combine into

$$\boxed{\frac{dP}{ds} = q \text{re}(FU) = \frac{q}{2}(FU + UF^H).} \quad (7.26)$$

Hermitian extraction in (7.26) enforces real mechanical momentum and reproduces the action-derived force. The imaginary part is the magnetic-dual force in Section 11.1; it need not vanish for an electric charge.

7.6 Acceleration and the mass shell

Recall $P = mU$, $U^*U = 1$, and real coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$. Scalar-part cyclicity proves tangency to the mass shell:

$$\begin{aligned} \text{sc}(U^* dP/ds) &= \frac{q}{2} \text{sc}(U^* F U + F^H) \\ &= \frac{q}{2} \left(\text{sc}(F) + \text{sc}(F^H) \right) = 0. \end{aligned}$$

This agrees with $\text{sc}(U^* dU/ds) = 0$ in Section 6.2. Combining (6.8) with the force components in (7.25) gives

$$\begin{aligned} m\gamma^3 \mathbf{v} \cdot \mathbf{a} &= q \mathbf{v} \cdot \mathbf{E}, \\ m\gamma (\mathbf{a} + \gamma^2 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v}) &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

Eliminating $\mathbf{v} \cdot \mathbf{a}$ yields the coordinate acceleration

$$\boxed{\mathbf{a} = \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E}) \mathbf{v})}. \quad (7.27)$$

Equation (7.27) shows how the magnetic term changes momentum without doing work. Taking the determinant of the real force gives

$$\begin{aligned} -\det\left(\frac{dP}{ds}\right) &= -m^2 \det\left(\frac{dU}{ds}\right) \\ &= q^2 \gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2). \end{aligned} \quad (7.28)$$

Dividing (7.28) by m^2 gives the invariant squared proper acceleration. The factor $\gamma^2((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2)$ is the squared electric field in the particle's instantaneous rest frame.

7.7 Energy and momentum flux

Recall $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Its Hermitian product is

$$\frac{1}{2} F F^H = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (7.29)$$

Define the real energy density $u = (\mathbf{E}^2 + \mathbf{B}^2)/2$ and real Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}$. The coefficients of (7.29) are frame-dependent, whereas F^2 contains the Lorentz invariants. The full stress-energy tensor is represented by the real linear map

$$T(N) := \frac{1}{2} F N F^H, \quad (7.30)$$

with real paravector input N . The time-direction value of (7.30) is $T(1) = u + \mathbf{S} \cdot \boldsymbol{\sigma}$. It is future causal because $u \geq 0$ and

$$\det(T(1)) = u^2 - \mathbf{S}^2 = \frac{1}{4} \|\det F\|^2 \geq 0. \quad (7.31)$$

Equality in (7.31) holds for null fields. Reversing the order gives $F^H F = (F F^H)^*$: the same energy density and opposite flux. The transformation of the full map is derived in Section 9.

Using Maxwell's vector equations,

$$\begin{aligned} \partial_t u &= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \\ &= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E}, \end{aligned}$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Thus Poynting's energy-balance law is

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.32)$$

Equivalently, the left side of (7.32) is $\frac{1}{2} \partial_t \text{sc}(FF^H) + \frac{1}{2} \partial_{\mathbf{r}} \cdot \text{vec}(FF^H)$. The right side is the field's loss of energy to matter per unit volume.

8 Relativistic wave equations

Throughout this section, $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ has real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$, $m \geq 0$ is mass, $q \in \mathbb{R}$ is charge, and $\hbar = c = 1$. Wave amplitudes are complex-valued functions of these real coordinates; their scalar, ideal, or two-entry type is specified below. Spin is fixed by the rotation law, not by the word “paravector.” We first identify the carriers and their probabilities, then derive the free and coupled equations before their low-energy limits and applications.

8.1 Spin and probability

8.1.1 Pairing and observables

Let ψ, ψ' be complex paravector amplitudes and Z a paravector acting from the left. Define

$$\langle \psi' | \psi \rangle := \text{sc}(\psi'^H \psi), \quad (8.1)$$

$$\langle \psi' | Z | \psi \rangle := \text{sc}(\psi'^H Z \psi). \quad (8.2)$$

The pairing (8.1) defines the real norm density

$$\rho := \text{sc}(\psi^H \psi) = \|\psi\|^2 \geq 0. \quad (8.3)$$

For a finite amplitude, (8.3) is its squared norm; for an ideal wavefunction it is the local density. Spatial normalization means $\int \rho \, d^3 \mathbf{r} = 1$, not $\psi^H \psi = 1$. The spatial norm is conserved by Pauli and, after summing both entries, Dirac; local densities may vary. The Klein–Gordon scalar uses a different conserved pairing.

For $\rho > 0$ and real $Z^H = Z$, normalize the matrix element (8.2) to define the mean

$$\langle Z \rangle_{\psi} := \frac{\text{sc}(\psi^H Z \psi)}{\rho}. \quad (8.4)$$

For a spatial wavefunction in (8.4), integrate numerator and denominator. A Hermitian differential operator can replace Z when its derivatives act to the right and boundary conditions justify integration by parts. The Born rule supplies the probability $\|\Pi \psi\|^2 / \rho$ for an orthogonal projector Π ; the algebra evaluates this physical postulate.

If $\psi' = T\psi \neq 0$ for a complex paravector T , then

$$\langle Z \rangle_{\psi'} = \frac{\text{sc}(\psi^H T^H Z T \psi)}{\text{sc}(\psi^H T^H T \psi)}. \quad (8.5)$$

Equation (8.5) shows that $T^H T = 1$ preserves the finite-amplitude norm. A Lorentz boost does not preserve this norm of spinor components: the invariant Dirac normalization is current flux through a spacelike hypersurface, including its transformed measure.

8.1.2 How the spin number is defined

Recall the real unit axis $\mathbf{u}$, real angle θ , and unitary rotor $R = \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.50). All intrinsic transformations below compare field values at corresponding rotated points. Rotating their coordinate dependence additionally produces orbital angular momentum.

Define the intrinsic spin operator $S(\mathbf{u})$ by writing an infinitesimal rotation as $1 - i\theta S(\mathbf{u}) + O(\theta^2)$ on the amplitude. Here $S(\mathbf{u})$ is a linear operator, whose action depends on the carrier. With the real Cartesian elementary vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, write $S_k = S(\mathbf{e}_k)$ and $S^2 = S_1^2 + S_2^2 + S_3^2$. The rotation algebra gives $[S_1, S_2] = iS_3$ and cyclic permutations. For spin number $s = 0, 1/2, 1, \dots$, an irreducible carrier has

$$S^2 = s(s+1). \quad (8.6)$$

Together with the rotation commutators, (8.6) gives projections $-s, -s+1, \dots, s$ along any axis, in units of $\hbar$.

8.1.3 Spin zero: a scalar amplitude

Let $\psi(\mathbf{r}, t) \in \mathbb{C}$ be a scalar field. Its intrinsic rotation law is

$$\psi' = \psi. \quad (8.7)$$

Equation (8.7) gives $S(\mathbf{u})\psi = 0$ for every axis and $S^2\psi = 0$, hence $s = 0$ by (8.6). There is one complex amplitude and only the zero intrinsic projection; spatial dependence can still carry orbital angular momentum.

A scalar KG field uses this law and stays scalar under boosts. The KG equation alone does not fix spin: free Dirac components satisfy the same equation but have the spinor rotation law.

8.1.4 Spin one-half: a one-sided ideal amplitude

For a complex ideal amplitude $\psi\Pi_3 = \psi$, with fixed $\Pi_3 = (1 + \sigma_3)/2$, the rotation acts from one side:

$$\psi' = R^H \psi. \quad (8.8)$$

Here $R = \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$. Expanding (8.8) identifies the generator; its square gives

$$S(\mathbf{u})\psi = \frac{1}{2}(\mathbf{u} \cdot \boldsymbol{\sigma})\psi, \quad (8.9)$$

$$S^2\psi = \frac{1}{4}(\sigma_1^2 + \sigma_2^2 + \sigma_3^2)\psi = \frac{3}{4}\psi. \quad (8.10)$$

Equations (8.6) and (8.10) give $s = 1/2$. Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, the generator (8.9) has projections $\pm 1/2$. A 2π rotation gives $R = -1$, reversing the amplitude's sign; a 4π rotation restores it. The sign cancels in probabilities but relative phases remain observable in interference.^[2]

8.1.5 Spin one: a two-sided vector amplitude

Let $\mathbf{V}(\mathbf{r}, t) \in \mathbb{C}^3$ be a complex vector field. With $R = \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ its intrinsic law is

$$\mathbf{V}' \cdot \boldsymbol{\sigma} = R^H (\mathbf{V} \cdot \boldsymbol{\sigma}) R. \quad (8.11)$$

Expanding both factors in (8.11) gives $\mathbf{V}' = \mathbf{V} + \theta \mathbf{u} \times \mathbf{V} + O(\theta^2)$, hence

$$S(\mathbf{u})\mathbf{V} = \mathbf{i}\mathbf{u} \times \mathbf{V}, \quad (8.12)$$

$$S^2\mathbf{V} = -\sum_{k=1}^3 \mathbf{e}_k \times (\mathbf{e}_k \times \mathbf{V}) = 2\mathbf{V}. \quad (8.13)$$

Equations (8.6) and (8.13) give $s = 1$. The third-axis eigenvectors of (8.12) satisfy

$$\begin{aligned} S_3\mathbf{e}_3 &= 0, \\ S_3(\mathbf{e}_1 \pm \mathbf{i}\mathbf{e}_2) &= \pm(\mathbf{e}_1 \pm \mathbf{i}\mathbf{e}_2). \end{aligned}$$

The projections are 0, +1, −1, and a 2π rotation restores the amplitude. A complex scalar plus this vector has separate spin-zero and spin-one rotation parts. A position or momentum paravector is a four-vector observable, not a particle's spin amplitude.

The electromagnetic field $F = (\mathbf{E} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}$ has this spin-one rotation law, with real fields $\mathbf{E}, \mathbf{B}$. Maxwell's equations constrain source-free radiation to be transverse. The physical photon has two helicities, +1 and −1, rather than three independent polarizations. A massive spin-one particle has three polarizations once its field equation and constraint are imposed. A classical field F is not itself a normalized particle amplitude.

The scalar, spinor, electromagnetic field, and four-vector also have distinct boost laws, derived in Section 9.

8.1.6 Spinor components and probabilities

Fix the real paravector $\Pi_3 = (1 + \sigma_3)/2$ on the right. For any complex paravector Z , set

$$\psi = Z\Pi_3. \quad (8.14)$$

Equation (8.14) imposes $\psi\Pi_3 = \psi$, preserved by every left action since $(Z'\psi)\Pi_3 = Z'\psi$. This minimal left ideal has two independent complex components. The fixed right projection selects one Pauli column; a measured spin outcome is a separate left projection. Z is not unique.

Choose the following two constant, normalized ideal amplitudes:

$$\{\psi_+, \psi_-\} := \sqrt{2}\{\Pi_3, \sigma_1\Pi_3\}. \quad (8.15)$$

The basis (8.15) obeys $\sigma_3\psi_{\pm} = \pm\psi_{\pm}$ and $\langle\psi_a | \psi_b\rangle = \delta_{ab}$ for $a, b \in \{+, -\}$. The scalar coefficients $c_+, c_- \in \mathbb{C}$ give the full expansion

$$\begin{aligned} \psi &= c_+\psi_+ + c_-\psi_- \\ &= \frac{c_+(1 + \sigma_3) + c_-(\sigma_1 - \mathbf{i}\sigma_2)}{\sqrt{2}}. \end{aligned} \quad (8.16)$$

Thus (8.3) becomes $\rho = \|c_+\|^2 + \|c_-\|^2$. For a wavefunction, $c_{\pm} = c_{\pm}(\mathbf{r}, t)$ are complex scalar functions. Equation (8.16) encodes four paravector coefficients using two independent complex components: the Pauli column entries in Appendix A.4. Since σ_1 exchanges the basis, neither one-dimensional subspace is invariant under all rotations.

For a real unit measurement axis $\mathbf{u}$, the real paravector projectors are

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (8.17)$$

The identities $\Pi_{\pm}^2 = \Pi_{\pm}$, $\Pi_+ \Pi_- = 0$, and $\Pi_+ + \Pi_- = 1$ for (8.17) imply

$$\begin{aligned}\psi &= \Pi_+ \psi + \Pi_- \psi, \\ \langle \Pi_+ \psi | \Pi_- \psi \rangle &= 0, \\ S(\mathbf{u})(\Pi_{\pm} \psi) &= \pm \frac{1}{2} \Pi_{\pm} \psi, \\ \|\Pi_{\pm} \psi\|^2 &= \text{sc}(\psi^H \Pi_{\pm} \psi).\end{aligned}$$

After a nonzero-probability outcome, the normalized amplitude is $\Pi_{\pm} \psi / \|\Pi_{\pm} \psi\|$. For the third axis, its probability is $\|c_{\pm}\|^2 / \rho$.

8.1.7 The real state paravector and probabilities

For $\rho > 0$, define the real paravector

$$B := \frac{\psi \psi^H}{\rho} = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}. \quad (8.18)$$

In (8.18), $\mathbf{s} = \text{vec}(B) \in \mathbb{R}^3$ is the dimensionless Bloch polarization; $\mathbf{s}/2$ is the mean spin. Since $\det \Pi_3 = 0$,

$$\det B = \frac{\|\det \psi\|^2}{\rho^2} = 1 - \mathbf{s}^2 = 0. \quad (8.19)$$

Equation (8.19) gives $\mathbf{s}^2 = 1$ for a nonzero ideal amplitude. A statistical mixture $B = \sum_k w_k B_k$ of these real pure-state paravectors, with real weights $w_k \geq 0$ and $\sum_k w_k = 1$, has $\mathbf{s}^2 \leq 1$. Conversely, every such real B can be factored as $\psi \psi^H / \rho$ using an unrestricted complex paravector. Its matrix image divided by two is the usual density matrix. The resulting ball of polarizations is state geometry: $\det B = 0$ means a pure spin state, not a massless particle.

Cyclicity of $\text{sc}(\cdot)$ gives

$$\begin{aligned}\|\Pi_{\pm} \psi\|^2 &= \text{sc}(\psi^H \Pi_{\pm} \psi) = \rho \text{sc}(\Pi_{\pm} B), \\ \text{sc}(\Pi_{\pm} B) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}).\end{aligned}$$

Hence the Born probabilities are

$$p_{\pm}(\mathbf{u}) = \frac{\|\Pi_{\pm} \psi\|^2}{\rho} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (8.20)$$

For a wavefunction, (8.20) is conditional on position. A detector collecting the whole packet measures

$$p_{\pm}(\mathbf{u}) = \frac{\int \text{sc}(\psi^H \Pi_{\pm} \psi) \, d^3 \mathbf{r}}{\int \rho \, d^3 \mathbf{r}}. \quad (8.21)$$

Equation (8.21) weights each local spin fraction by its density. A real observable $Z' = a + \mathbf{b} \cdot \boldsymbol{\sigma}$, with real scalar a and real vector $\mathbf{b}$, has possible values $a \pm \|\mathbf{b}\|$ and mean

$$\langle Z' \rangle_{\psi} = \text{sc}(Z' B) = a + \mathbf{b} \cdot \mathbf{s}. \quad (8.22)$$

For (8.22), use $\mathbf{u} = \mathbf{b} / \|\mathbf{b}\|$ when $\mathbf{b} \neq 0$ to obtain its outcome probabilities from (8.20).

The amplitude law (8.8), with $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, gives

$$B' = R^H B R. \quad (8.23)$$

Equation (8.23) rotates the polarization as a vector while the amplitude has spin one-half. An unrestricted paravector under the one-sided law contains two spin-one-half copies; the ideal selects one. Dirac couples two ideals with the same rotation law and opposite boost laws.

8.1.8 Spin uncertainty

For real unit axes $\mathbf{u}, \mathbf{u}'$, set $A = \mathbf{u} \cdot \boldsymbol{\sigma}/2$ and $A' = \mathbf{u}' \cdot \boldsymbol{\sigma}/2$. Define the real variance by $\text{var } A = \langle A^2 \rangle - \langle A \rangle^2$. The product rule gives

$$\begin{aligned} [A, A'] &= \frac{i}{2} (\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \text{var } A &= \frac{1 - (\mathbf{u} \cdot \mathbf{s})^2}{4}. \end{aligned}$$

For normalized ψ , positivity of $\|(A - \langle A \rangle)\psi + i\lambda(A' - \langle A' \rangle)\psi\|^2$ for every real auxiliary scalar λ makes its quadratic discriminant nonpositive. Therefore

$$\boxed{\text{var } A \text{ var } A' \geq \frac{1}{4} \|\langle [A, A'] \rangle\|^2 = \frac{[(\mathbf{u} \times \mathbf{u}') \cdot \mathbf{s}]^2}{16}.} \quad (8.24)$$

The Robertson bound (8.24) also holds for mixtures by the same positive-pairing argument.

8.2 Energy and momentum as differential operators

Recall $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$. For a plane wave let $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$ be a constant real four-momentum: $E \in \mathbb{R}$ is its energy and $\mathbf{p} \in \mathbb{R}^3$ its momentum. The real phase and complex scalar wave are

$$f(X) := \text{sc}(\text{adj}(P) X) = Et - \mathbf{p} \cdot \mathbf{r}, \quad (8.25)$$

$$w(X) := \exp(-if(X)). \quad (8.26)$$

Differentiating (8.26), with phase (8.25), gives the translation eigenvalues:

$$\begin{aligned} i\partial_t w &= Ew, \\ -i\partial_{\mathbf{r}} w &= \mathbf{p}w, \\ i\partial w &= \text{adj}(P) w. \end{aligned}$$

Their scalar and vector parts combine into one paravector equation,

$$i\partial^* w = Pw. \quad (8.27)$$

Equation (8.27) identifies $\text{sc}(i\partial^*) = i\partial_t$ and $\text{vec}(i\partial^*) = -i\partial_{\mathbf{r}}$ as the energy and momentum operators. Extraction here precedes their action on a wave. On the spatially integrated pairing, $-i\partial_{\mathbf{r}}$ is Hermitian when boundary terms vanish. An energy spectrum is real when the wave equation has a Hermitian Hamiltonian H in $i\partial_t \Psi = H\Psi$.

In a field, recall the real potential paravector $\Phi(t, \mathbf{r}) = V(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$: V is a real scalar potential and $\mathbf{A}$ a real vector potential. The real charge is q . The particle action gives canonical energy $E + qV$ and momentum $\mathbf{p} + q\mathbf{A}$; identifying them with translation operators defines the gauge-covariant mechanical operators

$$E := i\partial_t - qV, \quad (8.28)$$

$$\mathbf{p} := -i\partial_{\mathbf{r}} - q\mathbf{A}. \quad (8.29)$$

In (8.28), E includes rest energy; $i\partial_t$ is canonical energy. The operator (8.29) is mechanical momentum. Below, $E, \mathbf{p}$ denote these scalar and vector differential operators unless a real mode eigenvalue is specified. They act to the right and form

$$E - \mathbf{p} \cdot \boldsymbol{\sigma} = i\partial - q\Phi^*, \quad (8.30)$$

$$E + \mathbf{p} \cdot \boldsymbol{\sigma} = i\partial^* - q\Phi. \quad (8.31)$$

The components of (8.30) and (8.31) need not commute; their ordered product cannot be replaced by an ordinary paravector determinant. For free operators they commute, and $\square = \partial_t^2 - \partial_{\mathbf{r}}^2$ gives

$$(i\partial)(i\partial^*) = -\square. \quad (8.32)$$

Reversing the order in (8.32) gives the same result.

8.2.1 Field strength from noncommuting derivatives

Recall the real electric and magnetic fields from (7.12) and (7.13), and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real scalar in (7.11). The real potential is $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$; $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is a complex vector paravector. Apply the operators to a differentiable complex amplitude. The product rule cancels all derivatives of the amplitude in the commutators; for Cartesian indices $k, l \in \{1, 2, 3\}$,

$$[E, p_k] = -iq(\partial_t A_k + \partial_k V) = iqE_k,$$

$$[p_k, p_l] = iq(\partial_k A_l - \partial_l A_k),$$

$$\mathbf{p} \times \mathbf{p} = iq\mathbf{B}.$$

Sigma multiplication and the energy commutator consequently give

$$(\mathbf{p} \cdot \boldsymbol{\sigma})^2 = \mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (8.33)$$

$$[E, \mathbf{p} \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}. \quad (8.34)$$

Here E_k is an electric-field component. Using (8.33) and (8.34) in the ordered products gives

$$\begin{aligned} (E + \mathbf{p} \cdot \boldsymbol{\sigma})(E - \mathbf{p} \cdot \boldsymbol{\sigma}) &= E^2 - (\mathbf{p} \cdot \boldsymbol{\sigma})^2 - [E, \mathbf{p} \cdot \boldsymbol{\sigma}] \\ &= E^2 - \mathbf{p}^2 - iqF, \end{aligned} \quad (8.35)$$

$$\begin{aligned} (E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + \mathbf{p} \cdot \boldsymbol{\sigma}) &= E^2 - (\mathbf{p} \cdot \boldsymbol{\sigma})^2 + [E, \mathbf{p} \cdot \boldsymbol{\sigma}] \\ &= E^2 - \mathbf{p}^2 - iqF^*. \end{aligned} \quad (8.36)$$

Equations (8.35) and (8.36) share scalar part $E^2 - \mathbf{p}^2$; their vector remainders are $-iqF$ and $-iqF^*$. This separation will distinguish the scalar KG equation from the squared spinor equation.

8.3 Klein–Gordon equation

Let $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ be a scalar field and $m > 0$ its real mass. For a free real four-momentum, $\det P = m^2$ means $E^2 - \mathbf{p}^2 = m^2$. Replacing its coefficients by the translation operators in (8.27) gives

$$((i\partial)(i\partial^*) - m^2)\psi = 0.$$

Using (8.32), this is the Klein–Gordon equation:

$$\boxed{(\square + m^2)\psi = 0.} \quad (8.37)$$

8.3.1 Free solutions and initial data

To solve (8.37), define the positive spatial operator $H = \sqrt{m^2 - \partial_{\mathbf{r}}^2}$ by its Fourier symbol. Here $\mathbf{k} \in \mathbb{R}^3$ is the wavevector and $E(\mathbf{k}) > 0$ the mode energy, not a differential operator:

$$\begin{aligned} E(\mathbf{k}) &:= \sqrt{m^2 + \mathbf{k}^2}, \\ H \exp(i\mathbf{k} \cdot \mathbf{r}) &= E(\mathbf{k}) \exp(i\mathbf{k} \cdot \mathbf{r}). \end{aligned}$$

$$\square + m^2 = (\partial_t - iH)(\partial_t + iH). \quad (8.38)$$

By (8.38), each Fourier coefficient satisfies the oscillator equation $\partial_t^2 \psi + E(\mathbf{k})^2 \psi = 0$. Its two solutions give

$$\boxed{\psi(t, \mathbf{r}) = \int \frac{d^3 \mathbf{k}}{(2\pi)^{3/2}} \left(a(\mathbf{k}) e^{-iE(\mathbf{k})t} + b(\mathbf{k}) e^{iE(\mathbf{k})t} \right) e^{i\mathbf{k} \cdot \mathbf{r}}.} \quad (8.39)$$

In (8.39), the complex scalar functions $a, b : \mathbb{R}^3 \rightarrow \mathbb{C}$ are fixed by the initial field and its first time derivative. To display this directly, write $\psi = \psi_+ + \psi_-$, where the complex scalar fields $\psi_{\pm}$ obey

$$\begin{aligned} (\partial_t \pm iH)\psi_{\pm} &= 0, \\ \psi_{\pm}(0, \mathbf{r}) &= \frac{1}{2} (\psi(0, \mathbf{r}) \pm iH^{-1} \partial_t \psi(0, \mathbf{r})), \\ \psi_{\pm}(t, \mathbf{r}) &= e^{\mp iHt} \psi_{\pm}(0, \mathbf{r}). \end{aligned}$$

Equivalently, arbitrary initial data evolve by

$$\psi(t, \mathbf{r}) = \cos(Ht) \psi(0, \mathbf{r}) + H^{-1} \sin(Ht) \partial_t \psi(0, \mathbf{r}). \quad (8.40)$$

Equation (8.40) reconstructs the field from its value and time derivative. The square root acts by Fourier multiplication. The original $\sqrt{\square}$ factorization follows likewise, using real frequency ω :

$$\begin{aligned} \square e^{i(\omega t + \mathbf{k} \cdot \mathbf{r})} &= (\mathbf{k}^2 - \omega^2) e^{i(\omega t + \mathbf{k} \cdot \mathbf{r})}, \\ \square + m^2 &= (m - i\sqrt{\square})(m + i\sqrt{\square}). \end{aligned} \quad (8.41)$$

Choose one branch of $\sqrt{\mathbf{k}^2 - \omega^2}$ consistently in (8.41). This operator is distinct from the positive H ; the two on-shell frequencies are $\omega = \pm E(\mathbf{k})$ [4].

8.3.2 Potential and conserved current

Recall the real potentials $V, \mathbf{A}$, charge q , and kinetic operators $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The scalar equation in a field is

$$\boxed{(-E^2 + \mathbf{p}^2 + m^2)\psi = 0.} \quad (8.42)$$

Indeed, the scalar part of either ordered product $(E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + \mathbf{p} \cdot \boldsymbol{\sigma})$ is $E^2 - \mathbf{p}^2$. The vector remainder acts on spinor amplitudes in the Dirac factorization.

The product rule separately gives

$$E^2 = -\partial_t^2 - iq(\partial_t V) - 2iqV\partial_t + q^2V^2, \quad (8.43)$$

$$\mathbf{p}^2 = -\partial_{\mathbf{r}}^2 + iq(\partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2\mathbf{A}^2. \quad (8.44)$$

Substituting (8.43) and (8.44) into (8.42) gives

$$\boxed{(\square + iqS - q^2 \det \Phi + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}))\psi = 0.} \quad (8.45)$$

Equation (8.45) retains every derivative of the potential. Its real current is $J = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$, where $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}$ is its signed density and $\mathbf{J} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ its spatial flux:

$$J := \frac{1}{m} \operatorname{re}(\psi^H(E + \mathbf{p} \cdot \boldsymbol{\sigma})\psi). \quad (8.46)$$

Extracting (8.46) gives

$$\rho = \frac{i}{2m} \left(\psi^H \partial_t \psi - (\partial_t \psi)^H \psi \right) - \frac{qV}{m} \psi^H \psi, \quad (8.47)$$

$$\mathbf{J} = -\frac{i}{2m} \left(\psi^H \partial_{\mathbf{r}} \psi - (\partial_{\mathbf{r}} \psi)^H \psi \right) - \frac{q\mathbf{A}}{m} \psi^H \psi. \quad (8.48)$$

For the continuity check of (8.47) and (8.48), abbreviate the scalar wave operator as $Q = -E^2 + \mathbf{p}^2 + m^2$. The product rule cancels the products of first derivatives, leaving

$$\begin{aligned} \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} &= \frac{i}{2m} \left(\psi^H Q \psi - (Q \psi)^H \psi \right). \\ \operatorname{sc}(\partial J) &= \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0. \end{aligned} \quad (8.49)$$

To interpret the conserved current (8.49), take free waves $\psi = a \exp(\mp i \operatorname{sc}(\operatorname{adj}(P) X))$, where $a \in \mathbb{C}$ is constant and P has positive energy,

$$J = \pm \frac{\|a\|^2}{m} P. \quad (8.50)$$

Equation (8.50) identifies $q\rho$ as charge density; ρ is not a positive probability density for arbitrary KG fields. For the positive-frequency sector ($b = 0$ in (8.39)), Parseval's identity gives

$$\int \rho \, d^3 \mathbf{r} = \int \frac{E(\mathbf{k})}{m} \|a(\mathbf{k})\|^2 \, d^3 \mathbf{k} > 0 \quad (8.51)$$

for a nonzero field. The pairing (8.51) is positive even though the local density of a superposition need not be. A general time-dependent potential need not preserve this sector. The full particle–antiparticle interpretation requires the quantized scalar field; the positive Pauli and Dirac density cannot simply be assigned to a KG amplitude.

8.3.3 Schrödinger limit

Recall $m > 0$, real $V, \mathbf{A}, q$, and the kinetic operators $E, \mathbf{p}$ above. Let ψ' be the relativistic complex scalar field. Its envelope $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ removes the rest-energy phase:

$$\psi'(t, \mathbf{r}) = e^{-imt} \psi(t, \mathbf{r}). \quad (8.52)$$

Under (8.52), E acting on ψ measures energy above rest: $E\psi' = e^{-imt}(E + m)\psi$. Equation (8.42) becomes

$$\begin{aligned} (-E^2 - 2mE + \mathbf{p}^2)\psi &= 0, \\ E\psi &= \frac{\mathbf{p}^2 - E^2}{2m}\psi. \end{aligned}$$

On the free positive-frequency branch the commuting operators give

$$E\psi = \left(\sqrt{m^2 + \mathbf{p}^2} - m \right) \psi. \quad (8.53)$$

The small-momentum expansion of (8.53) is

$$E\psi = \left(\frac{\mathbf{p}^2}{2m} - \frac{\mathbf{p}^4}{8m^3} + \cdots \right) \psi. \quad (8.54)$$

More generally, when residual energies and momenta are small compared with m , dropping $E^2\psi$ at leading order gives Schrödinger's equation:

$$\boxed{\mathrm{i}\partial_t \psi = \left(\frac{(-\mathrm{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi.} \quad (8.55)$$

Equation (8.55) is the leading scalar limit. The higher terms of (8.54) assume commuting operators and cannot be imported unchanged into arbitrary time-dependent fields.

8.4 Dirac equation

8.4.1 Block form and free factorization

Recall $m > 0$, the real space-time coordinates $(t, \mathbf{r})$, and the fixed idempotent $\Pi_3 = (1 + \sigma_3)/2$. A Dirac field is the two-entry block

$$\Psi(t, \mathbf{r}) = \begin{Bmatrix} \Psi_1(t, \mathbf{r}) \\ \Psi_2(t, \mathbf{r}) \end{Bmatrix}. \quad (8.56)$$

Its entries satisfy $\Psi_k \Pi_3 = \Psi_k$ for $k = 1, 2$. Each complex ideal entry in (8.56) has two independent complex components, so the block has four. On blocks, $^{\mathrm{H}}$ transposes entries and applies the existing Hermitian operation to each; $*$ acts entrywise without transposing. For two such fields, Ψ and Ψ' , the local pairing is

$$\langle \Psi' \mid \Psi \rangle := \sum_{k=1}^2 \mathrm{sc}(\Psi_k'^{\mathrm{H}} \Psi_k). \quad (8.57)$$

The pairing (8.57) gives the positive real density

$$\|\Psi\|^2 = \mathrm{sc}(\Psi_1^{\mathrm{H}} \Psi_1) + \mathrm{sc}(\Psi_2^{\mathrm{H}} \Psi_2). \quad (8.58)$$

For an algebraic complex paravector Z , define the block map

$$\mathbf{W}(Z) := \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix}. \quad (8.59)$$

Here $\mathbf{I}$ is the two-entry block identity. Multiplying (8.59) gives

$$\mathbf{W}(Z)^2 = \begin{Bmatrix} Z \text{adj}(Z) & 0 \\ 0 & \text{adj}(Z) Z \end{Bmatrix} = \det(Z) \mathbf{I}. \quad (8.60)$$

Entrywise conjugation gives $\mathbf{W}(Z)^* = \mathbf{W}(Z^*)$; the Hermitian operation also transposes:

$$\mathbf{W}(Z)^H = \mathbf{W}(Z^*). \quad (8.61)$$

Equations (8.60) and (8.61) are algebraic identities. Differential Hermiticity additionally requires integration by parts; $\mathbf{W}(Z)$ is not generally Hermitian. For the free field define the operator blocks

$$\begin{aligned} \mathbf{D} &:= \mathbf{W}(\text{i}\partial) - m\mathbf{I}, \\ \mathbf{D}' &:= \mathbf{W}(\text{i}\partial) + m\mathbf{I}. \end{aligned}$$

The Dirac equation is

$$\boxed{\mathbf{D}\Psi = 0}. \quad (8.62)$$

Since $\mathbf{W}(\text{i}\partial)^2 = -\square\mathbf{I}$, its mass conjugate gives

$$\mathbf{D}'\mathbf{D} = \mathbf{W}(\text{i}\partial)^2 - m^2\mathbf{I} = -(\square + m^2)\mathbf{I}. \quad (8.63)$$

Applying (8.63) to (8.62) gives $(\square + m^2)\Psi = 0$: every Dirac component satisfies KG. The original first-order constraint is the coupled pair

$$\text{i}\partial\Psi_2 = m\Psi_1, \quad (8.64)$$

$$\text{i}\partial^*\Psi_1 = m\Psi_2. \quad (8.65)$$

8.4.2 Reconstructing free solutions from Klein–Gordon

Let $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$ solve KG. For constant complex scalars a, b , the original two-constant construction is

$$\Psi = \frac{1}{m} \begin{Bmatrix} ma + \text{i}b\partial \\ \text{i}a\partial^* + mb \end{Bmatrix} \psi' \Pi_3. \quad (8.66)$$

Substituting (8.66) into (8.64) and (8.65) gives

$$\begin{aligned} \text{i}\partial\Psi_2 - m\Psi_1 &= -\frac{a}{m}(\square + m^2)\psi'\Pi_3 = 0, \\ \text{i}\partial^*\Psi_1 - m\Psi_2 &= -\frac{b}{m}(\square + m^2)\psi'\Pi_3 = 0. \end{aligned}$$

Taking $a = 1, b = 0$ gives the scalar-seeded solution. One scalar seed with two constants does not span every Dirac field. For the complete converse, let $\psi : \mathbb{R}^4 \rightarrow \mathbb{C} + \mathbb{C}^3 \cdot \boldsymbol{\sigma}$ be an ideal-valued field ($\psi\Pi_3 = \psi$) whose two complex components each solve KG. Then

$$\boxed{\Psi = \begin{Bmatrix} \psi \\ \text{i}\partial^*\psi/m \end{Bmatrix}}. \quad (8.67)$$

In (8.67), the second Dirac equation holds by construction and the first reduces to KG. Conversely, every massive free solution has this form with $\psi = \Psi_1$. Applying (8.39) to its two coefficients therefore supplies arbitrary initial data.

For completeness, on a Fourier mode the ordinary 4×4 matrix of $\mathbf{D}$ has characteristic polynomial

$$\det(\text{mat}(\mathbf{D}) - sI_4) = ((s + m)^2 + \square)^2. \quad (8.68)$$

Here $s \in \mathbb{C}$ is the spectral parameter, I_4 the four-dimensional identity, and $\square$ denotes its scalar Fourier eigenvalue. Equation (8.68) has two roots $s = -m \pm i\sqrt{\square}$ each have multiplicity two; treating the paravector block as an ordinary scalar 2×2 matrix loses this factor.

8.4.3 Potential and paravector components

Recall the real $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, real charge q , and $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. Minimal coupling gives

$$\boxed{(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0.} \quad (8.69)$$

The entries of (8.69) are

$$(E - \mathbf{p} \cdot \boldsymbol{\sigma})\Psi_2 = m\Psi_1, \quad (8.70)$$

$$(E + \mathbf{p} \cdot \boldsymbol{\sigma})\Psi_1 = m\Psi_2. \quad (8.71)$$

For a real scalar gauge function $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$, the product rule preserves (8.70) and (8.71) under $\Phi' = \Phi + \partial^* \lambda$ and

$$\Psi' = e^{-iq\lambda}\Psi. \quad (8.72)$$

Equation (8.72) changes the local phase only. To expose each coefficient, write $\Psi_k = a_k + \mathbf{b}_k \cdot \boldsymbol{\sigma}$ for $k = 1, 2$, with complex scalar functions $a_k : \mathbb{R}^4 \rightarrow \mathbb{C}$ and complex vector functions $\mathbf{b}_k : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Sigma multiplication gives

$$Ea_2 - \mathbf{p} \cdot \mathbf{b}_2 = ma_1, \quad (8.73)$$

$$E\mathbf{b}_2 - \mathbf{p}a_2 - i\mathbf{p} \times \mathbf{b}_2 = m\mathbf{b}_1, \quad (8.74)$$

$$Ea_1 + \mathbf{p} \cdot \mathbf{b}_1 = ma_2, \quad (8.75)$$

$$E\mathbf{b}_1 + \mathbf{p}a_1 + i\mathbf{p} \times \mathbf{b}_1 = m\mathbf{b}_2. \quad (8.76)$$

The scalar equations (8.73), (8.75) and vector equations (8.74), (8.76) act on the coefficient to their right. The ideal condition imposes $b_{k3} = a_k$ and $b_{k2} = -ib_{k1}$; the independent coefficients are a_1, b_{11}, a_2, b_{21} . Their four complex equations are exactly the conventional Dirac system; Appendix A.4 supplies the column representation.

8.4.4 Hamiltonian, density, and flux

Isolating the time derivative gives $i\partial_t \Psi = H\Psi$, where the Hamiltonian block is

$$H := qV\mathbf{I} + \begin{Bmatrix} -\mathbf{p} \cdot \boldsymbol{\sigma} & m \\ m & \mathbf{p} \cdot \boldsymbol{\sigma} \end{Bmatrix}. \quad (8.77)$$

Equivalently $H = qV\mathbf{I} + \mathbf{W}(1)(m\mathbf{I} + \mathbf{W}(\mathbf{p} \cdot \boldsymbol{\sigma}))$. Real potentials and vanishing boundary terms make (8.77) Hermitian in the integrated pairing, so it conserves the norm (8.58).

Define the real probability-current paravector $J = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$ by

$$J := \text{adj}(\Psi_1 \Psi_1^H) + \Psi_2 \Psi_2^H. \quad (8.78)$$

Extracting (8.78) gives its nonnegative density $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}$ and spatial probability flux $\mathbf{J} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$:

$$\begin{aligned} \rho &= \text{sc}(\Psi_1^H \Psi_1) + \text{sc}(\Psi_2^H \Psi_2), \\ \mathbf{J} &= -\text{vec}(\Psi_1 \Psi_1^H) + \text{vec}(\Psi_2 \Psi_2^H). \end{aligned}$$

The component equations and their Hermitian forms give

$$\begin{aligned} \partial_t \|\Psi_1\|^2 - \partial_{\mathbf{r}} \cdot \text{vec}(\Psi_1 \Psi_1^H) &= 2m \text{im}(\text{sc}(\Psi_1^H \Psi_2)), \\ \partial_t \|\Psi_2\|^2 + \partial_{\mathbf{r}} \cdot \text{vec}(\Psi_2 \Psi_2^H) &= -2m \text{im}(\text{sc}(\Psi_1^H \Psi_2)). \end{aligned}$$

The mass transfers density between entries; summing cancels the transfer:

$$\text{sc}(\partial J) = \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0. \quad (8.79)$$

The conserved current (8.79) is a sum of two future-null terms by the ideal condition. Thus J is real and future causal, as the determinant confirms:

$$\det J = \rho^2 - \mathbf{J}^2 = 4 \left\| \text{sc}(\Psi_1^H \Psi_2) \right\|^2 \geq 0. \quad (8.80)$$

Equation (8.80) bounds the probability-flow velocity $\mathbf{J}/\rho$ by one wherever $\rho > 0$. Normalize by $\int \rho \, d^3\mathbf{r} = 1$. More generally, for a spacelike hypersurface let N be its future-pointing real unit normal ($\det N = 1$) and dA its real invariant volume element. The Lorentz-invariant normalization is

$$\int \text{sc}(\text{adj}(N) J) \, dA = 1. \quad (8.81)$$

On a constant-time surface $N = 1$, so (8.81) recovers the spatial integral.

8.4.5 Free Dirac modes and spin probabilities

Let $\mathbf{p} \in \mathbb{R}^3$ be fixed momentum and $m > 0$ the mass. Define the positive real energy $E = \sqrt{\mathbf{p}^2 + m^2}$. Set $V = \mathbf{A} = 0$ in the Hamiltonian (8.77) and replace the momentum operator by this mode's real $\mathbf{p}$. Squaring cancels the off-diagonal entries: $H^2 = (\mathbf{p}^2 + m^2)\mathbf{I} = E^2\mathbf{I}$, with $\mathbf{I}$ the block identity. Its orthogonal energy projectors are therefore

$$\mathbf{Q}_{\pm} := \frac{1}{2}(\mathbf{I} \pm H/E). \quad (8.82)$$

The projectors (8.82) obey $\mathbf{Q}_{\pm}^2 = \mathbf{Q}_{\pm} = \mathbf{Q}_{\pm}^H$, $\mathbf{Q}_+ \mathbf{Q}_- = 0$, and $\mathbf{Q}_+ + \mathbf{Q}_- = \mathbf{I}$. For a nonzero finite amplitude,

$$p_{\pm} = \frac{\|\mathbf{Q}_{\pm} \Psi\|^2}{\|\Psi\|^2}. \quad (8.83)$$

For a packet in (8.83), let $\Psi(\mathbf{p}, t)$ denote its spatial Fourier transform: apply $\mathbf{Q}_{\pm}(\mathbf{p})$ at each real momentum and integrate numerator and denominator over $d^3\mathbf{p}$. In position space $E = \sqrt{-\partial_{\mathbf{r}}^2 + m^2}$ makes the projectors nonlocal.

Constructing the normalized modes. At rest, $H = m\mathbf{W}(1)$, and the energy sign $a = \pm 1$ requires $\Psi_2 = a\Psi_1$. With the constant third-axis basis $\psi_\pm$ of (8.15), the four orthonormal rest amplitudes are

$$\Psi_{ab}(0) = \frac{1}{\sqrt{2}} \{\psi_b, a\psi_b\}^\top. \quad (8.84)$$

In (8.84), $a, b \in \{+1, -1\}$: a labels energy sign and $b/2$ spin along the chosen axis. Neither original entry by itself labels energy or spin.

For arbitrary momentum define the real paravector $A = E + m + \mathbf{p} \cdot \boldsymbol{\sigma}$. Its adjugate is $\text{adj}(A) = E + m - \mathbf{p} \cdot \boldsymbol{\sigma}$, and direct multiplication gives

$$\begin{aligned} A \text{adj}(A) &= (E + m)^2 - \mathbf{p}^2 = 2m(E + m), \\ A^2 + (\text{adj}(A))^2 &= 2[(E + m)^2 + \mathbf{p}^2] = 4E(E + m). \end{aligned}$$

The second identity and $A \text{adj}(A) = \text{adj}(A) A$ prove that the block map

$$\mathbf{T} := \frac{1}{2\sqrt{E(E + m)}} \begin{Bmatrix} \text{adj}(A) & A \\ A & -\text{adj}(A) \end{Bmatrix} \quad (8.85)$$

is Hermitian and unitary: in (8.85), $\mathbf{T}^\mathbf{H} = \mathbf{T}$ and $\mathbf{T}^2 = \mathbf{I}$. The eigenvalue equations follow component by component:

$$\begin{aligned} -(\mathbf{p} \cdot \boldsymbol{\sigma}) \text{adj}(A) + mA &= E \text{adj}(A), \\ m \text{adj}(A) + (\mathbf{p} \cdot \boldsymbol{\sigma}) A &= EA, \\ -(\mathbf{p} \cdot \boldsymbol{\sigma}) A - m \text{adj}(A) &= -EA, \\ mA - (\mathbf{p} \cdot \boldsymbol{\sigma}) \text{adj}(A) &= E \text{adj}(A). \end{aligned}$$

Thus the first column has energy $+E$ and the second has energy $-E$:

$$\mathbf{T}^\mathbf{H} H \mathbf{T} = \begin{Bmatrix} E & 0 \\ 0 & -E \end{Bmatrix}. \quad (8.86)$$

Equation (8.86) is the free Foldy–Wouthuysen change of representation in the chiral basis. It preserves positive norms and is distinct from a Lorentz boost.[5]

Now choose any real unit rest-spin axis $\mathbf{u}$. In the following formulas, ψ_b ($b = \pm 1$) denotes a constant normalized ideal amplitude with $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_b = b\psi_b$. Project the fixed $\sqrt{2}\Pi_3$ using $\Pi_b = (1 + b\mathbf{u} \cdot \boldsymbol{\sigma})/2$ and normalize. Its squared norm is $(1 + bu_3)/2$; when zero, project $\sqrt{2}\sigma_1\Pi_3$ instead. The resulting pair is orthonormal and recovers the third-axis basis at $\mathbf{u} = \mathbf{e}_3$.

The four normalized momentum modes are

$$\Psi_{+b}(\mathbf{p}) = \frac{\{\text{adj}(A) \psi_b, A\psi_b\}^\top}{2\sqrt{E(E + m)}}. \quad (8.87)$$

$$\Psi_{-b}(\mathbf{p}) = \frac{\{A\psi_b, -\text{adj}(A) \psi_b\}^\top}{2\sqrt{E(E + m)}}. \quad (8.88)$$

Unitarity of $\mathbf{T}$ proves normalization, orthogonality, and completeness of (8.87) and (8.88). Multiplying by $e^{-iaEt + i\mathbf{p} \cdot \mathbf{r}}$ gives the space-time mode, with the same spatial momentum for both energy signs. Hence every free packet is

$$\Psi(\mathbf{r}, t) = \sum_{a,b=\pm 1} \int \frac{d^3\mathbf{p}}{(2\pi)^{3/2}} c_{ab}(\mathbf{p}) \Psi_{ab}(\mathbf{p}) e^{-iaEt + i\mathbf{p} \cdot \mathbf{r}}. \quad (8.89)$$

Here $c_{ab} : \mathbb{R}^3 \rightarrow \mathbb{C}$ are scalar Fourier amplitudes. Spatial unit normalization is exactly

$$\sum_{a,b} \int \|c_{ab}(\mathbf{p})\|^2 d^3\mathbf{p} = 1. \quad (8.90)$$

Equation (8.90) normalizes the packet (8.89); each individual plane wave instead has unit finite spinor norm and an infinite spatial integral. The sum/difference basis in Appendix A.6 gives the familiar large and small components proportional to ψ_b and $(\mathbf{p} \cdot \boldsymbol{\sigma})\psi_b/(E + m)$.

For $m = 0$ and $\mathbf{p} \neq 0$, set $E = \|\mathbf{p}\|$ and $\mathbf{u} = \mathbf{p}/E$. The two chiral equations decouple. The complete normalized pair at energy aE is now simply

$$\{\psi_{-a}, 0\}^\top, \quad \{0, \psi_a\}^\top.$$

The first entry has helicity $-a/2$, the second $a/2$; helicity means $S(\mathbf{u})$ along momentum. At nonzero mass an entry has no definite helicity in general. For the exceptional $m = 0, \mathbf{p} = 0$ mode, $H = 0$: all four constant amplitudes have zero energy, without a preferred helicity or positive/negative split.

Rotation spin and conserved rest spin. For $\rho = \|\boldsymbol{\Psi}\|^2 > 0$, the rotation generator $\boldsymbol{\sigma}/2$ on each entry has the reduced real state paravector

$$B := \frac{\Psi_1 \Psi_1^H + \Psi_2 \Psi_2^H}{\rho} = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}. \quad (8.91)$$

In (8.91), $\mathbf{s}(\mathbf{r}, t) \in \mathbb{R}^3$ is the polarization and $\mathbf{s}/2$ the mean rotation spin. Summing over the two entries can give $\mathbf{s}^2 < 1$ even for a pure Dirac amplitude. The current J instead adjudicates its first term and encodes probability flow; neither reduced paravector specifies the full state.

For a real unit axis $\mathbf{u}$, let $\Pi_\pm = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ act on each entry. The local spin fractions are

$$\text{sc}(\Pi_\pm B) = \frac{\|\Pi_\pm \Psi_1\|^2 + \|\Pi_\pm \Psi_2\|^2}{\rho} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (8.92)$$

For a spatial packet, integrate the numerator and denominator of (8.92). At finite momentum the rotation generator differs from conserved rest spin; a detector's interaction determines which operator it measures.

At each fixed momentum put $\boldsymbol{\Psi}' = \mathbf{T}^H \boldsymbol{\Psi}$. Here $\boldsymbol{\Psi}'$ is the same finite amplitude in the diagonal representation, and $\rho = \|\boldsymbol{\Psi}\|^2 = \|\boldsymbol{\Psi}'\|^2$ is its real norm. Energy sign is now the entry label. The rest-spin operator is $\mathbf{u} \cdot \boldsymbol{\sigma}/2$ on either entry, so its projectors in the original representation are

$$\Pi'_b := \mathbf{T} \begin{Bmatrix} \Pi_b & 0 \\ 0 & \Pi_b \end{Bmatrix} \mathbf{T}^H. \quad (8.93)$$

The projectors (8.93) obey $\Pi_b'^2 = \Pi_b' = \Pi_b'^H$ and $[H, \Pi_b'] = 0$. Unitary conjugation preserves the Pauli commutators and squared spin $3/4$. This is the conserved Foldy–Wouthuysen mean spin for rotation-free rest axes; the construction here assumes $m > 0$. [7]

Since energy and rest-spin projectors commute, their four joint probabilities are

$$p_{ab} = \frac{\|Q_a \Pi'_b \boldsymbol{\Psi}\|^2}{\rho}. \quad (8.94)$$

In the diagonal representation, (8.94) becomes

$$\begin{aligned} p_{+b} &= \frac{\|\Pi_b \Psi'_1\|^2}{\rho}, \\ p_{-b} &= \frac{\|\Pi_b \Psi'_2\|^2}{\rho}, \\ \sum_{a,b} p_{ab} &= 1. \end{aligned}$$

For a wave packet, integrate numerator and denominator over momentum. The normalized expansion (8.89) gives joint weights $\int \|c_{ab}(\mathbf{p})\|^2 d^3\mathbf{p}$ directly.

At a fixed momentum the real rest-spin state paravector is

$$B' = \frac{\Psi'_1 \Psi'^H_1 + \Psi'_2 \Psi'^H_2}{\rho} = 1 + \mathbf{s}' \cdot \boldsymbol{\sigma}, \quad (8.95)$$

In (8.95), $\mathbf{s}' \in \mathbb{R}^3$ is its rest polarization. Summing over energy gives $p_b = (1 + b\mathbf{u} \cdot \mathbf{s}')/2$. For a pure energy eigenmode, use $(\mathbf{p} \cdot \boldsymbol{\sigma})(\mathbf{s}' \cdot \boldsymbol{\sigma})(\mathbf{p} \cdot \boldsymbol{\sigma}) = 2(\mathbf{p} \cdot \mathbf{s}')\mathbf{p} \cdot \boldsymbol{\sigma} - \mathbf{p}^2 \mathbf{s}' \cdot \boldsymbol{\sigma}$ in the two mode products. Their linear cross terms cancel, giving

$$\mathbf{s} = \frac{m}{E} \mathbf{s}' + \frac{\mathbf{p}(\mathbf{p} \cdot \mathbf{s}')}{E(E + m)}. \quad (8.96)$$

Equation (8.96) leaves parallel polarization unchanged; transverse polarization in B is reduced by m/E . The full state remains pure and its spin remains one-half. Only the reduced rotation-spin state appears mixed.

The untransformed Π_b instead measures the Dirac rotation generator. Acting on both entries, its commutator with energy is

$$[H, \Pi_b] = \begin{Bmatrix} -ib(\mathbf{p} \times \mathbf{u}) \cdot \boldsymbol{\sigma} & 0 \\ 0 & ib(\mathbf{p} \times \mathbf{u}) \cdot \boldsymbol{\sigma} \end{Bmatrix}. \quad (8.97)$$

The commutator (8.97) vanishes at rest or along $\mathbf{p}$; there Π_b equals the rest-spin projector. Along nonzero momentum it selects helicity. Otherwise $\mathbf{Q}_a \Pi_b$ is not a joint orthogonal projector. For ideal sequential measurements, let $p(a, b)$ mean energy a followed by rotation spin b , and $p'(a, b)$ the reverse order:

$$p(a, b) = \frac{\|\Pi_b \mathbf{Q}_a \Psi\|^2}{\rho}, \quad (8.98)$$

$$p'(a, b) = \frac{\|\mathbf{Q}_a \Pi_b \Psi\|^2}{\rho}. \quad (8.99)$$

The joint probabilities (8.98) and (8.99) need not agree. Conditioning (8.98) on a nonzero energy outcome gives

$$p(b \mid a) = \frac{\|\Pi_b \mathbf{Q}_a \Psi\|^2}{\|\mathbf{Q}_a \Psi\|^2}. \quad (8.100)$$

Equation (8.100) is the joint probability divided by the probability of the first outcome. The slow Stern–Gerlach experiment of Section 8.5.1 measures the ordinary Pauli component in its stated approximation.

Covariant idempotents and charge reversal. Recall the future real momentum $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, real mass $m > 0$, and real unit-determinant paravector $U = P/m$. The real phase is $f(\mathbf{r}, t) = Et - \mathbf{p} \cdot \mathbf{r}$. Since $\det U = 1$, the block $\mathbf{W}(\text{adj}(U))$ squares to $\mathbf{I}$. Consequently $(\mathbf{I} \pm \mathbf{W}(\text{adj}(U)))/2$ are idempotents. Their $+$ eigenspace solves the Dirac equation with phase e^{-if} ; the $-$ eigenspace uses e^{+if} , with spatial momentum $-\mathbf{p}$. The latter corresponds to $\mathbf{Q}_-(-\mathbf{p})$. Away from rest,

$$\mathbf{W}(\text{adj}(U))^H = \mathbf{W}(U) \neq \mathbf{W}(\text{adj}(U)). \quad (8.101)$$

Equation (8.101) shows why these idempotents are not Hermitian away from rest. Born probabilities use squared norms of the orthogonal $\mathbf{Q}_\pm$ projections, divided by the initial squared norm.

For a charge- q solution, define its charge-conjugate complex ideal block using a prime locally:

$$\Psi' := \{\Psi_2^* \sigma_1, -\Psi_1^* \sigma_1\}^T. \quad (8.102)$$

Equation (8.102) solves the charge- $(-q)$ equation. Indeed, $\Pi_3^* \sigma_1 = \sigma_1 \Pi_3$ keeps both entries in the fixed ideal; applying $*$ to the coupled equations reverses i and the sign of the charge coupling. It also reverses the phase, while $\Pi_\pm^* = \Pi_\mp$ reverses the rotation-spin label. A negative-frequency charge- q mode thereby becomes a positive-frequency charge- $(-q)$ mode, with spin evaluated on the resulting amplitude. After field quantization the original negative-frequency coefficients become antiparticle creation amplitudes. Negative energy by itself is not an antimatter probability.

Proper orthochronous Lorentz transformations preserve the two free mass shells. A static external field instead uses the full Hermitian Hamiltonian's spectrum. A time-dependent field need not preserve a fixed energy split; a pair-production interpretation requires field quantization.

8.4.6 Interacting factorization and reconstruction

Recall the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, the complex field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. The operators $E, \mathbf{p}$ and complex ideal entries of Ψ retain their preceding meanings. The scalar part of their ordered product gave KG; Dirac retains its vector spin coupling as well.

For any complex paravector amplitude $\psi(t, \mathbf{r})$ the same calculation can be kept in paravector form:

$$\partial^*(\Phi^* \psi) + \Phi \partial \psi = (S + F) \psi + 2(V \partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \psi, \quad (8.103)$$

$$\partial(\Phi \psi) + \Phi^* \partial^* \psi = (S + F^*) \psi + 2(V \partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \psi. \quad (8.104)$$

The difference of (8.103) and (8.104) is $(F - F^*) \psi = 2(\mathbf{E} \cdot \boldsymbol{\sigma}) \psi$.

For the block map $\mathbf{W}$, direct multiplication gives

$$\mathbf{W}(\partial) \mathbf{W}(\Phi^*) = \begin{Bmatrix} \partial \Phi & 0 \\ 0 & \partial^* \Phi^* \end{Bmatrix},$$

$$\mathbf{W}(\Phi^*) \mathbf{W}(\partial) = \begin{Bmatrix} \Phi^* \partial^* & 0 \\ 0 & \Phi \partial \end{Bmatrix},$$

$$\mathbf{W}(\Phi^*)^2 = \det \Phi \mathbf{I}.$$

Here $\partial \Phi$ denotes composition: on a test field it is $\partial(\Phi \psi)$, including the product rule. Define the interacting mass-conjugate blocks

$$\mathbf{D} := \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I},$$

$$\mathbf{D}' := \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}.$$

Multiplying and using the preceding mixed-derivative identities gives

$$\mathbf{D}'\mathbf{D} = - \begin{Bmatrix} -E^2 + \mathbf{p}^2 + m^2 + iqF^* & 0 \\ 0 & -E^2 + \mathbf{p}^2 + m^2 + iqF \end{Bmatrix}. \quad (8.105)$$

Equation (8.105) recovers (8.63) at $q = 0$. With coupling, its entries obey

$$\begin{aligned} ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2) \Psi_1 &= 0, \\ ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2) \Psi_2 &= 0. \end{aligned}$$

Expanding gives the exact second-order equations

$$(-E^2 + \mathbf{p}^2 + m^2 + iqF^*)\Psi_1 = 0, \quad (8.106)$$

$$(-E^2 + \mathbf{p}^2 + m^2 + iqF)\Psi_2 = 0. \quad (8.107)$$

The spin terms in (8.106) and (8.107) also have an explicit coefficient check. For this calculation only, write $\mathbf{C} = \mathbf{E} + i\mathbf{B}$ and $\psi = a + \mathbf{b} \cdot \boldsymbol{\sigma}$, where $a : \mathbb{R}^4 \rightarrow \mathbb{C}$ and $\mathbf{b}, \mathbf{C} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$ are complex coefficient functions. The star on $\mathbf{C}$ conjugates its coefficients. Then

$$iqF\psi = iq\mathbf{C} \cdot \mathbf{b} + (iq\mathbf{C} - q\mathbf{C} \times \mathbf{b}) \cdot \boldsymbol{\sigma}, \quad (8.108)$$

$$iqF^*\psi = -iq\mathbf{C}^* \cdot \mathbf{b} + (-iq\mathbf{C}^* + q\mathbf{C}^* \times \mathbf{b}) \cdot \boldsymbol{\sigma}. \quad (8.109)$$

Equations (8.108) and (8.109) display the scalar and vector couplings separately.

To construct solutions, take a complex ideal field ψ satisfying $(-E^2 + \mathbf{p}^2 + m^2 + iqF)\psi = 0$ and $\psi\Pi_3 = \psi$. Set

$$\boxed{\Psi = \begin{Bmatrix} (E - \mathbf{p} \cdot \boldsymbol{\sigma})\psi/m \\ \psi \end{Bmatrix}}. \quad (8.110)$$

In (8.110), the first Dirac equation holds by construction; the second is the seed equation. Every massive solution has this form with $\psi = \Psi_2$. Independent squared entries need not satisfy their first-order coupling, and a scalar KG seed generally lacks the required iqF term. Arbitrary potentials still require solving their initial or boundary problem; this formula reconstructs Dirac from the correct seed.

Closing the matter–field system. Let J be the real Dirac current (8.78) and $D = qJ$ the real electric source paravector. With $F^* = \partial\Phi - \text{sc}(\partial\Phi)$, the coupled Maxwell equation is

$$\partial F = (qJ)^*. \quad (8.111)$$

Combining (8.111) with (8.69) gives $\text{sc}(\partial D) = q\text{sc}(\partial J) = 0$, exactly the source consistency condition. Replacing the Dirac current by (8.46) gives Maxwell–KG. These are classical field systems; photons, particle creation, and radiative corrections require quantum field theory.

8.4.7 Schrödinger–Pauli limit

Recall $m > 0$, real charge q , real potentials $V, \mathbf{A}$, and the kinetic operators $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The complex field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ enters through (8.33) and (8.34). Remove the rest phase of the positive-energy Dirac field by defining complex ideal envelopes $\Psi'_k(t, \mathbf{r})$ for $k = 1, 2$:

$$\Psi'_k := e^{imt}\Psi_k. \quad (8.112)$$

The product rule gives

$$\begin{aligned} E\Psi_k &= e^{-imt}(E+m)\Psi'_k, \\ \mathbf{p}\Psi_k &= e^{-imt}\mathbf{p}\Psi'_k. \end{aligned}$$

On the envelopes (8.112), E measures energy above rest. Cancelling the common phase gives

$$\begin{aligned} (E+m-\mathbf{p}\cdot\boldsymbol{\sigma})\Psi'_2 &= m\Psi'_1, \\ (E+m+\mathbf{p}\cdot\boldsymbol{\sigma})\Psi'_1 &= m\Psi'_2. \end{aligned}$$

Eliminating Ψ'_1 retains both parts of F :

$$(E^2 + 2mE - \mathbf{p}^2 + q\mathbf{B}\cdot\boldsymbol{\sigma} - iq\mathbf{E}\cdot\boldsymbol{\sigma})\Psi'_2 = 0. \quad (8.113)$$

The imaginary electric term belongs to the second-order equation (8.113); Ψ'_2 alone is not a normalized first-order Pauli state.

Define the complex ideal amplitudes ϕ, χ by the sum and difference of the envelopes:

$$\phi := \frac{\Psi'_1 + \Psi'_2}{\sqrt{2}}, \quad (8.114)$$

$$\chi := \frac{\Psi'_2 - \Psi'_1}{\sqrt{2}}. \quad (8.115)$$

Substituting (8.114) and (8.115) gives the exact pair

$$E\phi = (\mathbf{p}\cdot\boldsymbol{\sigma})\chi, \quad (8.116)$$

$$(E+2m)\chi = (\mathbf{p}\cdot\boldsymbol{\sigma})\phi. \quad (8.117)$$

The norm is $\int (\|\phi\|^2 + \|\chi\|^2) d^3\mathbf{r}$. For residual energies and momenta small compared with m , (8.116) and (8.117) give

$$\begin{aligned} \chi &\simeq \frac{\mathbf{p}\cdot\boldsymbol{\sigma}}{2m}\phi, \\ E\phi &\simeq \frac{(\mathbf{p}\cdot\boldsymbol{\sigma})^2}{2m}\phi. \end{aligned}$$

Thus χ is small and the leading amplitude ϕ obeys Pauli's equation:

$$\boxed{i\partial_t\phi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B}\cdot\boldsymbol{\sigma} \right)\phi.} \quad (8.118)$$

In (8.118), $\phi\Pi_3 = \phi$ and $\int \text{sc}(\phi^H\phi) d^3\mathbf{r} = 1$ at leading order: ϕ is the Pauli amplitude denoted by ψ in the spin discussion. Its two complex coefficients in the normalized ideal basis ψ_+, ψ_- are precisely the conventional Pauli column. Since spin is $\boldsymbol{\sigma}/2$, the magnetic energy is $-(q/m)\mathbf{B}\cdot(\boldsymbol{\sigma}/2)$: the magnetic moment is q/m times spin, and the gyromagnetic factor is 2 before radiative corrections.

For a nonzero static uniform field, define the real unit vector $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$ and projectors $\Pi_{\pm} = (1 \pm \mathbf{u}\cdot\boldsymbol{\sigma})/2$. The projected ideal fields $\phi_{\pm} = \Pi_{\pm}\phi$ obey separate equations:

$$i\partial_t\phi_{\pm} = \left(\frac{\mathbf{p}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right)\phi_{\pm}. \quad (8.119)$$

The separation (8.119) uses a fixed direction. For a varying direction the projectors are functions of $(t, \mathbf{r})$. Writing H for the Pauli Hamiltonian in (8.118), differentiation gives

$$\begin{aligned} i\partial_t(\Pi_{\pm}\phi) &= i(\partial_t\Pi_{\pm})\phi + \Pi_{\pm}H\phi \\ &= H(\Pi_{\pm}\phi) + \left(i\partial_t\Pi_{\pm} + \frac{[\Pi_{\pm}, \mathbf{p}^2]}{2m}\right)\phi. \end{aligned} \quad (8.120)$$

In (8.120), only projector derivatives and the kinetic commutator couple the sectors; the magnetic term commutes with its own projectors.

8.4.8 First relativistic corrections

Recall the complex ideal amplitudes ϕ, χ in (8.117), real $m > 0, q, V, \mathbf{A}$, and $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. Assume static potentials and a positive-energy state separated from the negative-energy sector. Use a small real counting parameter v with

$$\mathbf{p}/m = O(v), \quad qV/m = O(v^2), \quad q\mathbf{B}/m^2 = O(v^2).$$

Spatial variation on scales $1/(mv)$ or longer then gives $q\mathbf{E}/m^2 = O(v^3)$. This is a smooth-field low-velocity expansion.

Iteration of the small-component equation preserves operator order:

$$\begin{aligned} \chi &= \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})\phi}{2m} - \frac{E(\mathbf{p} \cdot \boldsymbol{\sigma})\phi}{4m^2} + O(v^5\phi), \\ E\phi &= \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2\phi}{2m} - \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})E(\mathbf{p} \cdot \boldsymbol{\sigma})\phi}{4m^2} + O(mv^6\phi). \end{aligned}$$

For static potentials, $E(\mathbf{p} \cdot \boldsymbol{\sigma}) = (\mathbf{p} \cdot \boldsymbol{\sigma})E + [\mathbf{p} \cdot \boldsymbol{\sigma}, qV]$. Using the leading $E\phi = (\mathbf{p} \cdot \boldsymbol{\sigma})^2\phi/(2m)$ gives

$$E\phi = \left(\frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2}{2m} - \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^4}{8m^3} - \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})[\mathbf{p} \cdot \boldsymbol{\sigma}, qV]}{4m^2} \right) \phi + O(mv^6\phi). \quad (8.121)$$

The last term in (8.121) appears non-Hermitian because ϕ alone does not yet carry the full conserved norm:

$$\int (\|\phi\|^2 + \|\chi\|^2) d^3\mathbf{r} = \int \text{sc}(\phi^H \left(1 + \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2}{4m^2}\right) \phi) d^3\mathbf{r} + O(v^4). \quad (8.122)$$

Define the normalized complex ideal amplitude

$$\phi' := \left(1 + \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2}{8m^2}\right) \phi + O(v^4\phi). \quad (8.123)$$

Equation (8.123) absorbs the norm correction (8.122). Conjugating the evolution operator adds $[(\mathbf{p} \cdot \boldsymbol{\sigma})^2, qV]/(8m^2)$, combining the electric terms into a Hermitian double commutator. In $i\partial_t\phi' = H\phi'$,

$$\begin{aligned} H &= qV + \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2}{2m} - \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^4}{8m^3} \\ &\quad - \frac{[\mathbf{p} \cdot \boldsymbol{\sigma}, [\mathbf{p} \cdot \boldsymbol{\sigma}, qV]]}{8m^2} + O(mv^6). \end{aligned} \quad (8.124)$$

Equation (8.124) is the static Foldy–Wouthuysen positive-energy Hamiltonian through order mv^4 [5]. Direct sigma multiplication gives

$$[\mathbf{p} \cdot \boldsymbol{\sigma}, qV] = iq\mathbf{E} \cdot \boldsymbol{\sigma}.$$

$$[\mathbf{p} \cdot \boldsymbol{\sigma}, [\mathbf{p} \cdot \boldsymbol{\sigma}, qV]] = q\partial_{\mathbf{r}} \cdot \mathbf{E} + q(\mathbf{E} \times \mathbf{p} - \mathbf{p} \times \mathbf{E}) \cdot \boldsymbol{\sigma}. \quad (8.125)$$

The ordered cross products in (8.125) act on everything to their right; their difference is Hermitian. The explicit result is

$$\boxed{H = qV + \frac{\mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} - \frac{(\mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})^2}{8m^3} - \frac{q}{8m^2}\partial_{\mathbf{r}} \cdot \mathbf{E} - \frac{q}{8m^2}(\mathbf{E} \times \mathbf{p} - \mathbf{p} \times \mathbf{E}) \cdot \boldsymbol{\sigma} + O(mv^6)}. \quad (8.126)$$

The square in (8.126) retains the order of all noncommuting factors. For $\mathbf{A} = 0$, the three corrections are

$$\begin{aligned} \text{relativistic kinetic energy:} & \quad -\frac{\mathbf{p}^4}{8m^3}, \\ \text{Darwin term:} & \quad \frac{q}{8m^2}\partial_{\mathbf{r}}^2 V, \\ \text{spin-orbit term:} & \quad \frac{q}{4m^2}(\partial_{\mathbf{r}} V \times \mathbf{p}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

For a real central potential $V(r)$, with $r = \|\mathbf{r}\|$, the orbital operator $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ and spin operator $\mathbf{S} = \boldsymbol{\sigma}/2$ put the last term in its familiar form

$$\frac{q}{4m^2}(\partial_{\mathbf{r}} V \times \mathbf{p}) \cdot \boldsymbol{\sigma} = \frac{q}{2m^2 r} \frac{dV}{dr} \mathbf{L} \cdot \mathbf{S}. \quad (8.127)$$

Equation (8.127) includes the relativistic factor of one half. The gyromagnetic factor remains 2 to this order; its anomaly requires radiative corrections. The imaginary electric part of the exact squared Dirac equation has become Hermitian electric-field corrections after transforming both the state and its norm.

8.5 Worked solutions

8.5.1 Stern–Gerlach separation and spin precession

Let $m > 0$ be an atom's mass and $\mu \in \mathbb{R}$ its signed magnetic-moment coefficient: within an isolated spin-one-half doublet its moment operator is $\mu\boldsymbol{\sigma}$. Let $\psi(\mathbf{r}, t)\Pi_3 = \psi(\mathbf{r}, t)$ be its complex ideal wavefunction and $\mathbf{B}(\mathbf{r}) \in \mathbb{R}^3$ the magnetic field. The effective Pauli Hamiltonian is the Hermitian differential operator

$$H = -\frac{\partial_{\mathbf{r}}^2}{2m} - \mu\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (8.128)$$

With (8.128), slow center-of-mass motion obeys $i\partial_t\psi = H\psi$. Other internal levels are neglected. A neutral atom can have $\mu \neq 0$ because its constituents carry magnetic moments. For an elementary Dirac charge, equation (8.118) gives $\mu = q/(2m)$ with orbital electromagnetic coupling.

Use a local vacuum field with real gradient G and bias $B_0 > 0$:

$$\mathbf{B} = \{-Gx, 0, B_0 + Gz\}. \quad (8.129)$$

Here $\mathbf{r} = \{x, y, z\}$. Both the divergence and curl of (8.129) vanish. For a beam along y assume $|Gx|, |Gz| \ll B_0$ in the magnet. The spin axis is then approximately fixed along z , while its longitudinal gradient supplies the force. If the direction varies along a trajectory, the additional adiabatic condition $\|\mathbf{d}(\mathbf{B}/\|\mathbf{B}\|)/dt\| \ll 2|\mu|B_0$ suppresses transitions between its local spin states.

Let the initial wavefunction be $\psi(\mathbf{r}, 0) = f(\mathbf{r})\psi_0$, where $f : \mathbb{R}^3 \rightarrow \mathbb{C}$ has unit spatial norm and ψ_0 is a constant ideal amplitude with $\|\psi_0\| = 1$. With $\Pi_{\pm} = (1 \pm \sigma_3)/2$, write

$$\psi(\mathbf{r}, t) = f_+(\mathbf{r}, t)\Pi_+\psi_0 + f_-(\mathbf{r}, t)\Pi_-\psi_0. \quad (8.130)$$

In (8.130), the complex scalar functions $f_{\pm} : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C}$ each start from f . Substitution and $\sigma_3\Pi_{\pm} = \pm\Pi_{\pm}$ give

$$i\partial_t f_{\pm} = \left(-\frac{\partial_{\mathbf{r}}^2}{2m} \mp \mu B_0 \mp \mu Gz \right) f_{\pm}. \quad (8.131)$$

Let $f'(\mathbf{r}, t)$ solve the free scalar Schrödinger equation with the same initial function. To solve (8.131), translate its argument to cancel the derivative term from a linear phase; the remaining constant term fixes the cubic phase. Explicit substitution gives

$$f_{\pm}(\mathbf{r}, t) = \exp\left(\pm i\mu B_0 t \pm i\mu Gtz - \frac{i\mu^2 G^2 t^3}{6m}\right) f'\left(x, y, z \mp \frac{\mu Gt^2}{2m}, t\right). \quad (8.132)$$

Equation (8.132) gives momentum kicks $\pm\mu Gt$ and center displacements are $\pm\mu Gt^2/(2m)$ during transit. Subsequent free flight adds $\pm\mu GtT/m$ for flight time T . Each scalar packet retains unit norm. With the initial real state paravector $B = \psi_0\psi_0^H = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}$, resolved spots have

$$p_{\pm} = \|\Pi_{\pm}\psi_0\|^2 = \frac{1}{2}(1 \pm s_3). \quad (8.133)$$

The probabilities (8.133) depend on the initial spin; the sign of μG decides which label bends upward. After selecting spin $+1/2$ along $\mathbf{u}$, an analyzer along $\mathbf{u}'$ gives $(1 \pm \mathbf{u} \cdot \mathbf{u}')/2$, or $\cos^2(\theta/2)$ and $\sin^2(\theta/2)$ at mutual angle θ .^[3]

For a uniform field $\mathbf{B} = B_0\mathbf{u}$, there is no separation. Factoring out free spatial motion leaves the exact spin solution

$$\begin{aligned} \psi(t) &= \exp(i\mu B_0 t \mathbf{u} \cdot \boldsymbol{\sigma})\psi(0) \\ &= [\cos(\mu B_0 t) + i\mathbf{u} \cdot \boldsymbol{\sigma} \sin(\mu B_0 t)]\psi(0). \end{aligned} \quad (8.134)$$

Using (8.134) to differentiate $B = \psi\psi^H/\rho$ gives $\partial_t B = i\mu[\mathbf{B} \cdot \boldsymbol{\sigma}, B]$, whose vector part is

$$\partial_t \mathbf{s} = 2\mu \mathbf{s} \times \mathbf{B}. \quad (8.135)$$

For initial polarization along x and field along z , (8.135) gives

$$\mathbf{s}(t) = \{\cos(2\mu B_0 t), -\sin(2\mu B_0 t), 0\}. \quad (8.136)$$

An x analyzer applied to (8.136) therefore gives $p_+(t) = \cos^2(\mu B_0 t)$.

8.5.2 Exact solutions in constant potentials

Let $V \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$ be constant potentials, with real $m > 0, q$. Their field is $F = 0$. For real canonical frequency ω and wavevector $\mathbf{k} \in \mathbb{R}^3$, take the complex scalar mode

$$w(t, \mathbf{r}) = e^{-i\omega t + i\mathbf{k} \cdot \mathbf{r}}.$$

Its real kinetic momentum is $\mathbf{p} = \mathbf{k} - q\mathbf{A}$; the kinetic operators act as

$$\begin{aligned} (i\partial_t - qV)w &= (\omega - qV)w, \\ (-i\partial_{\mathbf{r}} - q\mathbf{A})w &= \mathbf{p}w. \end{aligned}$$

The mass shell gives $(\omega - qV)^2 = m^2 + \mathbf{p}^2$, hence both frequencies:

$$\boxed{\omega = qV \pm \sqrt{m^2 + (\mathbf{k} - q\mathbf{A})^2}} \quad (8.137)$$

In (8.137), ω is canonical energy; $\omega - qV$ is mechanical energy, including rest energy. Fourier superposition with arbitrary complex coefficients gives every KG solution, as in (8.39). For Dirac, each branch has the two spin modes derived in Section 8.4.5, with real kinetic momentum $\mathbf{k} - q\mathbf{A}$. Completeness also follows directly by gauge transformation. On simply connected space set the real function $\lambda = -Vt + \mathbf{A} \cdot \mathbf{r}$. The primed potential $\Phi' = \Phi + \partial^* \lambda$ vanishes, and arbitrary free fields ψ', Ψ' give

$$\psi(t, \mathbf{r}) = e^{-iqVt + iq\mathbf{A} \cdot \mathbf{r}} \psi'(t, \mathbf{r}), \quad (8.138)$$

$$\Psi(t, \mathbf{r}) = e^{-iqVt + iq\mathbf{A} \cdot \mathbf{r}} \Psi'(t, \mathbf{r}). \quad (8.139)$$

The maps (8.138) and (8.139) are invertible and preserve currents and probabilities. Constant potentials shift canonical labels and phases; they leave the kinetic mass shell unchanged.

8.5.3 Exact magnetic levels: Klein–Gordon, Dirac, and Pauli

Let $\mathbf{r} = \{x, y, z\} \in \mathbb{R}^3$ and prescribe the real uniform magnetic field $\mathbf{B} = \{0, 0, B\}$ with $B > 0$. Choose real potentials $\mathbf{A} = \{0, Bx, 0\}$ and constant $V = V_0$. Take real $m > 0, q \neq 0$, and define the positive scalar $d = |qB|$. The real labels k_2, k_3 are canonical momenta along y, z . On a mode with factor $e^{i(k_2y + k_3z)}$, the kinetic momentum square is

$$\mathbf{p}^2 = -\partial_x^2 + (k_2 - qBx)^2 + k_3^2.$$

Define the real dimensionless coordinate $u = \sqrt{d}(x - k_2/(qB))$. Then

$$\mathbf{p}^2 = d(-\partial_u^2 + u^2) + k_3^2. \quad (8.140)$$

Oscillator functions and scalar modes. The oscillator in (8.140) has $[\partial_u + u, u - \partial_u] = 2$. This gives the real normalized functions $f_N : \mathbb{R} \rightarrow \mathbb{R}$, with integer $N = 0, 1, 2, \dots$:

$$f_0(u) = \pi^{-1/4} e^{-u^2/2},$$

$$f_N(u) = \frac{(u - \partial_u)^N}{\sqrt{2^N N!}} f_0(u). \quad (8.141)$$

Raising the index in (8.141) preserves $\int |f_N(u)|^2 du = 1$ and gives

$$(-\partial_u^2 + u^2)f_N = (2N + 1)f_N. \quad (8.142)$$

For fixed N, k_2, k_3 , define the complex scalar spatial mode

$$w(\mathbf{r}) := \frac{d^{1/4}}{2\pi} f_N(u) e^{i(k_2y + k_3z)}. \quad (8.143)$$

Equation (8.142) gives $\mathbf{p}^2 w = (k_3^2 + d(2N + 1))w$. The mode (8.143) is normalized in x and delta-normalized in k_2, k_3 . The positive real KG mode energy is

$$E = \sqrt{m^2 + k_3^2 + d(2N + 1)}. \quad (8.144)$$

Using (8.144) in (8.42) gives both exact modes:

$$\boxed{\psi(t, \mathbf{r}) = e^{-i(qV_0 \pm E)t} w(\mathbf{r}).} \quad (8.145)$$

Multiplying (8.145) by $\sqrt{m/E}$ gives unit magnitude of the KG pairing for a normalized discrete mode (or unit delta normalization here). Its sign is the sign of the mechanical frequency $\pm E$, regardless of the shift qV_0 of the canonical frequency.

Dirac modes and all four components. For $s = \pm 1$, let ψ_s be a constant complex ideal spin amplitude with $\sigma_3 \psi_s = s \psi_s$ and $\|\psi_s\|^2 = 1$. Explicitly, $\psi_+ = \sqrt{2} \Pi_3$ and $\psi_- = \sqrt{2} \sigma_1 \Pi_3$. Define the ideal spatial seed $a(\mathbf{r}) = w(\mathbf{r}) \psi_s$. Sigma multiplication gives

$$\begin{aligned} (\mathbf{p} \cdot \boldsymbol{\sigma})^2 &= \mathbf{p}^2 - qB\sigma_3, \\ (\mathbf{p} \cdot \boldsymbol{\sigma})^2 a &= (k_3^2 + d(2N+1) - qBs)a. \end{aligned}$$

For these spinor modes the positive real energy is

$$E := \sqrt{m^2 + k_3^2 + d(2N+1) - qBs}. \quad (8.146)$$

The energy (8.146) gives two exact Dirac blocks:

$$\Psi_+(t, \mathbf{r}) = \frac{e^{-i(qV_0+E)t}}{2\sqrt{E(E+m)}} \begin{Bmatrix} (E+m - \mathbf{p} \cdot \boldsymbol{\sigma})a \\ (E+m + \mathbf{p} \cdot \boldsymbol{\sigma})a \end{Bmatrix}. \quad (8.147)$$

$$\Psi_-(t, \mathbf{r}) = \frac{e^{-i(qV_0-E)t}}{2\sqrt{E(E+m)}} \begin{Bmatrix} (E+m + \mathbf{p} \cdot \boldsymbol{\sigma})a \\ -(E+m - \mathbf{p} \cdot \boldsymbol{\sigma})a \end{Bmatrix}. \quad (8.148)$$

For (8.147), the first component equation follows from

$$(E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + m + \mathbf{p} \cdot \boldsymbol{\sigma})a = m(E + m - \mathbf{p} \cdot \boldsymbol{\sigma})a,$$

since $(\mathbf{p} \cdot \boldsymbol{\sigma})^2 a = (E^2 - m^2)a$. The other component and the negative branch (8.148) follow by the same multiplication. Hermiticity under spatial integration gives each block the seed's normalization because

$$\frac{(E+m)^2 + E^2 - m^2}{2E(E+m)} = 1.$$

No division by $E - m$ occurs, so this includes the lowest level. Every spatial component is explicit:

$$(\mathbf{p} \cdot \boldsymbol{\sigma})a = \frac{d^{1/4} e^{i(k_2 y + k_3 z)}}{2\pi} \left(-i\sqrt{d} \sigma_1 \partial_u - \frac{qB}{\sqrt{d}} u \sigma_2 + k_3 \sigma_3 \right) f_N(u) \psi_s. \quad (8.149)$$

The derivatives in (8.149) are adjacent oscillator functions:

$$\begin{aligned} u f_N &= \frac{\sqrt{N+1} f_{N+1} + \sqrt{N} f_{N-1}}{\sqrt{2}}, \\ \partial_u f_N &= \frac{\sqrt{N} f_{N-1} - \sqrt{N+1} f_{N+1}}{\sqrt{2}}. \end{aligned}$$

For $N = 0$ the term proportional to $\sqrt{N}$ vanishes. These are the usual four-component Dirac magnetic solutions [6].

Degeneracy and the Pauli limit. Define the combined nonnegative integer $N' = N + (1 - \text{sgn}(qB)s)/2$. The energy levels are

$$\boxed{E^2 = m^2 + k_3^2 + 2dN'}. \quad (8.150)$$

In (8.150), the lowest level $N' = 0$ has only $N = 0$ and $s = \text{sgn}(qB)$; each higher level has two spin-oscillator pairs. The seed's $s/2$ is spin in the representation separating positive and negative energies. Through $\mathbf{p} \cdot \boldsymbol{\sigma}$, the full Dirac block mixes its raw σ_3 components; it need not be an eigenstate of σ_3 acting on both entries.

Expanding the positive branch gives the residual energy

$$E - m = \frac{k_3^2 + d(2N + 1) - qBs}{2m} - \frac{(k_3^2 + d(2N + 1) - qBs)^2}{8m^3} + \dots. \quad (8.151)$$

Equation (8.151) gives orbital spacing $|qB|/m$ and Zeeman shift $-qBs/(2m)$: the Pauli result for spin $s/2$ and gyromagnetic factor 2. The next term agrees with (8.126) when $\mathbf{E} = 0$. Summing over N, s , integrating over k_2, k_3 , and including both energy branches spans arbitrary initial data in this field.

9 Rotations and Lorentz boosts

We now apply the algebraic actions of Section 4 to the physical variables already defined. Primed fields are evaluated at X' and unprimed fields at X . Factors are constant until the final subsection.

9.1 Covariance of the physical equations

Let T be a complex paravector with $\det T = 1$. It may be any ordered product of $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real unit $\mathbf{u}$ and real angle or rapidity θ . Recall the real event X , momentum P , potential Φ , electric current D , and Dirac probability current J . Each transforms by

$$Z' = T^{-1}Z(T^{-1})^H, \quad Z \in \{X, P, \Phi, D, J\}. \quad (9.1)$$

The derivative, complex field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, and complex ideal Dirac entries Ψ_1, Ψ_2 use the corresponding laws

$$\partial' = T^H \partial T, \quad (9.2)$$

$$F' = T^{-1}FT, \quad (9.3)$$

$$\Psi'_1 = T^H \Psi_1, \quad (9.4)$$

$$\Psi'_2 = T^{-1} \Psi_2. \quad (9.5)$$

Equation (9.1) acts on real four-vectors; (9.2) is their real-coordinate chain rule. The field similarity (9.3) and chiral laws (9.4)–(9.5) are fixed by covariance. Since $T^* = (T^H)^{-1}$, Maxwell covariance follows by cancellation:

$$\partial' F' = T^H (\partial F) T = T^H D^* T = D'^*. \quad (9.6)$$

Equation (9.6) recovers (7.2). For real charge q and mass $m > 0$, the coupled Dirac equations transform as

$$(i\partial' - q\Phi'^*)\Psi'_2 = T^H(i\partial - q\Phi^*)\Psi_2 = m\Psi'_1, \quad (9.7)$$

$$(i\partial'^* - q\Phi')\Psi'_1 = T^{-1}(i\partial^* - q\Phi)\Psi_1 = m\Psi'_2. \quad (9.8)$$

Equations (9.7) and (9.8) verify the opposite chiral factors. A single ideal amplitude needs a specified chirality before it has a boost law. The current law follows from (8.78), since

$$\text{adj}(\Psi'_1 \Psi_1^{\text{H}}) = T^{-1} \text{adj}(\Psi_1 \Psi_1^{\text{H}}) (T^{-1})^{\text{H}},$$

and the second current term has the same congruence.

Adjugation of P' and scalar-part cyclicity give

$$\text{sc}(\text{adj}(P') X') = \text{sc}(T^{\text{H}} \text{adj}(P) X (T^{\text{H}})^{-1}) = \text{sc}(\text{adj}(P) X). \quad (9.9)$$

Equation (9.9) makes $Et - \mathbf{p} \cdot \mathbf{r}$ an invariant wave phase. Replacing P, X by a real future normal N with $\det N = 1$ and current J gives the invariant flux $\text{sc}(\text{adj}(N) J)$ in (8.81). These are different actions on different physical objects, all within the same algebra.

9.2 Rotations and measurement probabilities

Recall $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real angle θ and real unit axis $\mathbf{u}$. Since $R^{\text{H}} = R^{-1}$, the preceding laws give $X' = R^{\text{H}} X R$ and $F' = R^{\text{H}} F R$. Time is unchanged; Rodrigues' formula (4.11) rotates the position and both field vectors.

For a nonzero complex ideal amplitude ψ and real observable Z ,

$$\begin{aligned} \psi' &= R^{\text{H}} \psi, \\ Z' &= R^{\text{H}} Z R, \\ \psi'^{\text{H}} Z' \psi' &= \psi^{\text{H}} Z \psi, \\ \psi'^{\text{H}} \psi' &= \psi^{\text{H}} \psi. \end{aligned}$$

The expectation definition (8.4) therefore gives

$$\langle Z' \rangle_{\psi'} = \langle Z \rangle_{\psi}. \quad (9.10)$$

Equation (9.10) includes projector probabilities. Both Dirac entries have this same one-sided rotation, and their current has the two-sided action. At 2π , amplitudes change sign while quadratic observables remain unchanged. The column translation is in Appendix A.4.

9.3 Moving frames, clock rates, and lengths

Recall $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real rapidity θ , real unit $\mathbf{u}$, $\beta = \tanh \theta$, and $\gamma = \cosh \theta$. The explicit gamma form is (3.66). For real $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, the passive boost $X' = L^{-1} X L^{-1}$ means that the primed frame moves at $+\beta \mathbf{u}$. Equations (4.13)–(4.14) become

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (9.11)$$

$$\mathbf{r}' = \mathbf{r}_{\perp} + \gamma(\mathbf{r}_{\parallel} - \beta t \mathbf{u}). \quad (9.12)$$

where $\mathbf{r}_{\parallel} = (\mathbf{u} \cdot \mathbf{r}) \mathbf{u}$ and $\mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}$. Reversing θ gives $X = L X' L$. For a clock at rest in the primed frame, (9.11) uses $d\mathbf{r} = \beta \mathbf{u} dt$, hence

$$dt' = \gamma(1 - \beta^2) dt = dt/\gamma. \quad (9.13)$$

Equation (9.13) is the moving clock rate. For a rod at rest in the unprimed frame, let $\Delta \mathbf{r}$ be its endpoint separation. A simultaneous measurement in the primed frame requires $\Delta t' = 0$, so $\Delta t = \beta \mathbf{u} \cdot \Delta \mathbf{r}$. Substitution gives

$$\Delta \mathbf{r}'_{\parallel} = \Delta \mathbf{r}_{\parallel} / \gamma. \quad (9.14)$$

The transverse separation remains unchanged by (9.12). Equation (9.14) is longitudinal contraction measured at equal primed times.

Collinear boosts add rapidities by (3.70). For a second real rapidity θ' , set $\beta' = \tanh \theta'$ and $\gamma' = \cosh \theta'$. Their combined parameters are

$$\beta'' = \tanh(\theta + \theta') = \frac{\beta + \beta'}{1 + \beta\beta'}, \quad (9.15)$$

$$\gamma'' = \gamma\gamma'(1 + \beta\beta'). \quad (9.16)$$

Equations (9.15) and (9.16) are the collinear velocity and clock-factor composition laws. Noncollinear boosts instead have the rotation commutator (3.71), producing the Thomas–Wigner rotation.

9.4 Boosted fields and energy flux

For $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma} / 2)$ with the parameters above, $F' = L^{-1}FL$. Apply (4.15) to $\mathbf{E} + i\mathbf{B}$ and separate its real and imaginary parts:

$$\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E}_{\perp} + \beta \mathbf{u} \times \mathbf{B}), \quad (9.17)$$

$$\mathbf{B}' = \mathbf{B}_{\parallel} + \gamma(\mathbf{B}_{\perp} - \beta \mathbf{u} \times \mathbf{E}). \quad (9.18)$$

Equations (9.17) and (9.18) leave parallel components unchanged; their projections use the same unit axis as (9.12). The general laws give the chiral boosts $\Psi'_1 = L\Psi_1$ and $\Psi'_2 = L^{-1}\Psi_2$, together with $\partial' = L\partial L$ and $\partial'^* = L^{-1}\partial^*L^{-1}$.

Recall the stress map $T(N) = FN F^H / 2$, with real paravector input N . Since $N' = L^{-1}NL^{-1}$,

$$T'(N') = \frac{1}{2}F'N'F'^H = L^{-1}T(N)L^{-1}. \quad (9.19)$$

Here T' denotes the map formed from F' . Equation (9.19) shows that each energy–momentum flux has the real four-vector law; $T(1)$ alone is only the flux through the chosen time direction. The same cancellation in (7.26) gives the transformed proper force.

9.5 Varying frames and local bases

Let $T(s)$ be a smooth complex paravector of determinant one along a real parameter s . Define the complex paravector $\Omega(s) := T^{-1}dT/ds$. The inverse rule gives

$$\frac{dT^{-1}}{ds} = -\Omega T^{-1}.$$

For a complex paravector $Z(s)$ transformed by similarity, $Z' = T^{-1}ZT$, product differentiation yields

$$\frac{dZ'}{ds} = T^{-1} \frac{dZ}{ds} T + [Z', \Omega]. \quad (9.20)$$

Differentiating (9.20) again and replacing $T^{-1}(\mathrm{d}Z/\mathrm{d}s)T$ by its first-derivative expression gives

$$\begin{aligned}\frac{\mathrm{d}^2 Z'}{\mathrm{d}s^2} &= T^{-1} \frac{\mathrm{d}^2 Z}{\mathrm{d}s^2} T + 2 \left[\frac{\mathrm{d}Z'}{\mathrm{d}s}, \Omega \right] \\ &\quad + \left[Z', \frac{\mathrm{d}\Omega}{\mathrm{d}s} \right] + [\Omega, [Z', \Omega]].\end{aligned}\tag{9.21}$$

The last term in (9.21) follows from $-[[Z', \Omega], \Omega] = [\Omega, [Z', \Omega]]$. For a real paravector transformed instead by congruence, $Z' = T^{-1}Z(T^{-1})^H$, the same rule gives

$$\frac{\mathrm{d}Z'}{\mathrm{d}s} = T^{-1} \frac{\mathrm{d}Z}{\mathrm{d}s} (T^{-1})^H - \Omega Z' - Z' \Omega^H.\tag{9.22}$$

Equations (9.20) and (9.22) include the frame terms omitted by a constant-factor transformation. For a varying rotation $T = R = \exp(i\theta(s)\mathbf{u}(s) \cdot \boldsymbol{\sigma}/2)$, with real $\theta(s)$ and real unit $\mathbf{u}(s)$, unitarity gives $\Omega^H = -\Omega$, so both formulas agree.

When s is proper time, applying the appropriate formula to the pointwise transformed tangent U gives its acceleration law. This differs from first transforming the event X and then differentiating: a varying event map already contributes frame terms to its first derivative. The similarity formula also gives the local algebra-basis law $\sigma'_k = T^{-1}\sigma_k T$, with

$$\frac{\mathrm{d}\sigma'_k}{\mathrm{d}s} = [\sigma'_k, \Omega], \quad k = 1, 2, 3.\tag{9.23}$$

The evolution (9.23) preserves all generator products. A general complex local basis is nevertheless distinct from a real spacetime frame, whose vectors require the congruence law.

10 Flat spacetime in generalized coordinates

Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{g} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be smooth functions on a coordinate chart with real coordinates $(t, \mathbf{r})$. They specify inertial coordinates through the real paravector

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

Require the Jacobian of $(f, \mathbf{g})$ to be nonsingular. Differentiating gives

$$\begin{aligned}\mathrm{d}X &= (\partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}) \mathrm{d}t + \sum_{k=1}^3 (\partial_{r_k} f + \partial_{r_k} \mathbf{g} \cdot \boldsymbol{\sigma}) \mathrm{d}r_k, \\ \mathrm{d}s^2 &= \mathrm{d}X \mathrm{d}X^* = \mathrm{d}f^2 - \mathrm{d}\mathbf{g}^2.\end{aligned}$$

With $x_0 = t$, $x_k = r_k$ and $\partial_k = \partial/\partial x_k$, the interval is $\mathrm{d}s^2 = \sum_{j,k=0}^3 g_{jk} \mathrm{d}x_j \mathrm{d}x_k$, with real coefficients

$$g_{jk} = \partial_j f \partial_k f - \partial_j \mathbf{g} \cdot \partial_k \mathbf{g}, \quad j, k \in \{0, 1, 2, 3\}.\tag{10.1}$$

Equation (10.1) is the flat Minkowski metric in changed coordinates; varying coefficients do not imply curvature.

For a future timelike curve, s is proper time and $\mathrm{d}f > 0$. Define its real inertial three-velocity $\mathbf{v} = \mathrm{d}\mathbf{g}/\mathrm{d}f$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, with $\mathbf{v}^2 < 1$. Then

$$\begin{aligned}\mathrm{d}s &= \sqrt{1 - \mathbf{v}^2} \mathrm{d}f, \\ U &= \frac{\mathrm{d}X}{\mathrm{d}s} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).\end{aligned}$$

Apply (6.8) with inertial time f and coordinate acceleration $d\mathbf{v}/df$. For inertial motion, the scalar part gives $\mathbf{v} \cdot d\mathbf{v}/df = 0$, and its vector part then gives $d\mathbf{v}/df = 0$. Therefore

$$\mathbf{g} = \mathbf{C}_1 f + \mathbf{C}_2. \quad (10.2)$$

The real constants $\mathbf{C}_1, \mathbf{C}_2 \in \mathbb{R}^3$ in (10.2) are velocity and initial position. If primes denote total derivatives along this curve with respect to t , then, wherever $f' \neq 0$,

$$\begin{aligned} \frac{d\mathbf{v}}{dt} &= \frac{\mathbf{g}'' f' - \mathbf{g}' f''}{(f')^2}, \\ [U, dU/ds] &= 2i\gamma^2 \left(\mathbf{v} \times \frac{d\mathbf{v}}{ds} \right) \cdot \boldsymbol{\sigma}. \end{aligned}$$

11 Magnetic duality and constant-field motion

This is the standard classical magnetic-source extension on real spacetime. Its magnetic charge is a hypothetical source parameter; no observed monopole is assumed. Setting it to zero recovers ordinary electrodynamics.

11.1 The imaginary force and magnetic charge

Recall the real proper velocity $U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ and momentum $P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, with real constant $m > 0$ and real proper time s . The real and imaginary parts of the product (7.25) are the electric force and its magnetic dual per unit charge. Let $q, q' \in \mathbb{R}$ be electric and magnetic charges; here the prime labels an independent charge. The classical dyon force is

$$\boxed{\frac{dP}{ds} = \text{re}((q - iq')FU)}. \quad (11.1)$$

The real components of (11.1) are

$$\frac{dE}{dt} = (q\mathbf{E} + q'\mathbf{B}) \cdot \mathbf{v}, \quad (11.2)$$

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) + q'(\mathbf{B} - \mathbf{v} \times \mathbf{E}). \quad (11.3)$$

Equation (11.2) is the work rate; (11.3) contains the electric and magnetic forces. The minus sign in $q - iq'$ follows from $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$: a positive magnetic charge feels $+q'\mathbf{B}$ at rest. Both forces preserve the mass shell, since cyclicity and $UU^* = 1$ give

$$\text{sc}(U^*FU) = \text{sc}(FUU^*) = \text{sc}(F) = 0.$$

Its real and imaginary parts give the two tangency conditions separately.

Let $M = d + \mathbf{M} \cdot \boldsymbol{\sigma}$ be a real magnetic-current paravector, with real magnetic charge density $d(t, \mathbf{r})$ and current $\mathbf{M}(t, \mathbf{r}) \in \mathbb{R}^3$. Maxwell's extension is

$$\boxed{\partial F = D^* + iM^*}. \quad (11.4)$$

Equation (11.4) retains the electric-source laws (7.3), (7.5), and (7.8). Its magnetic parts become

$$\partial_{\mathbf{r}} \cdot \mathbf{B} = d, \quad (11.5)$$

$$\partial_{\mathbf{r}} \times \mathbf{E} + \partial_t \mathbf{B} = -\mathbf{M}, \quad (11.6)$$

$$\partial_t d + \partial_{\mathbf{r}} \cdot \mathbf{M} = 0. \quad (11.7)$$

The divergence and curl in (11.5) and (11.6) imply magnetic-charge conservation (11.7). For any real constant angle a , the classical equations admit the simultaneous duality transformation

$$F \mapsto e^{-ia} F, \quad (11.8)$$

$$q + iq' \mapsto e^{-ia}(q + iq'), \quad (11.9)$$

$$D + iM \mapsto e^{-ia}(D + iM). \quad (11.10)$$

Equations (11.8) and (11.10) multiply both sides of Maxwell's law by the same phase; (11.9) leaves $(q - iq')F$ unchanged. A specified quantum charge spectrum need not admit every angle. Since a smooth curl has zero divergence, magnetic charge requires patched, singular, or enlarged potentials.[8] This extension uses real spacetime.

11.2 Exact motion in a constant field

Recall real charges q, q' , mass $m > 0$, and complex electromagnetic field F . For constant F , define

$$T(s) = \exp\left(\frac{(q - iq')s}{2m} F\right). \quad (11.11)$$

Using (11.11), the solution for future real initial momentum $P(0)$ with $\det P(0) = m^2$ is

$$P(s) = T(s)P(0)T(s)^H. \quad (11.12)$$

Differentiating (11.12), with real $U = P/m$,

$$\begin{aligned} \frac{dP}{ds} &= \frac{(q - iq')FP + P((q - iq')F)^H}{2m} \\ &= \text{re}((q - iq')FU). \end{aligned}$$

This congruence preserves Hermiticity, determinant, and the future cone; $\det T = 1$ follows from $\text{sc}(F) = 0$. Hence $U = P/m$ remains a physical proper velocity. For $q' = 0$, real unit $\mathbf{u}$, and constant real field magnitudes E, B ,

$$\begin{aligned} T(s) &= \exp(iqBs\mathbf{u} \cdot \boldsymbol{\sigma}/(2m)) & (\mathbf{E} = 0, \mathbf{B} = B\mathbf{u}), \\ T(s) &= \exp(qEs\mathbf{u} \cdot \boldsymbol{\sigma}/(2m)) & (\mathbf{B} = 0, \mathbf{E} = E\mathbf{u}). \end{aligned}$$

These are $R = \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ with $\theta = qBs/m$ and $L = \exp(\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ with $\theta = qEs/m$. The evolution uses the inverse sandwiches to (4.10) and (4.12) for the same factors. Complex field factors thus already generate real motion.

12 Extrapolation and speculation

The following coordinate and force laws are additional hypotheses. Lorentz covariance, the ordinary force law, stability, and the cosmological stress tensor provide established tests of these toy models. The tests do not identify a dark sector or an electroweak interaction with imaginary coordinates.

12.1 Testing a complex momentum equation

Keep real charges q, q' and electromagnetic field F as above, but postulate a complex paravector $P(s)$ with

$$\frac{dP}{ds} = (q + iq')FU. \quad (12.1)$$

Initially let s and U remain real. Extracting (12.1) gives

$$\begin{aligned} \frac{d \operatorname{re}(P)}{ds} &= q \operatorname{re}(FU) - q' \operatorname{im}(FU), \\ \frac{d \operatorname{im}(P)}{ds} &= q \operatorname{im}(FU) + q' \operatorname{re}(FU). \end{aligned}$$

The real magnetic force has the opposite sign to (11.1). Even when $q' = 0$, the imaginary force $q \operatorname{im}(FU)$ remains. At rest in a pure magnetic field, $FU = i\mathbf{B} \cdot \boldsymbol{\sigma}$: the proposed law produces imaginary momentum where an electric charge stays at rest. Thus $P = mU$ cannot remain real in general. If P is independent of the real trajectory, these equations only accumulate two impulses; they do not prescribe a complex trajectory.

To close the model, impose $U = P/m$ with real $m > 0$, allowing both paravectors to be complex. Now $s \in \mathbb{R}$ is an evolution parameter, not established physical proper time. The law becomes

$$\frac{dP}{ds} = \frac{q + iq'}{m} FP.$$

It preserves the complex determinant because

$$\begin{aligned} \frac{d \det P}{ds} &= 2 \operatorname{sc}(\operatorname{adj}(P) \frac{dP}{ds}) \\ &= \frac{2(q + iq')}{m} \det P \operatorname{sc}(F) = 0. \end{aligned}$$

For constant F , the complete solution is

$$P(s) = \exp\left(\frac{(q + iq')s}{m} F\right) P(0). \quad (12.2)$$

Take $q' = 0$, $q \neq 0$, $\mathbf{E} = 0$, constant $\mathbf{B} = \{0, 0, B\}$ with real $B \neq 0$, and $P(0) = m$. Equation (12.2) gives

$$P(s) = m \cos(qBs/m) + im\sigma_3 \sin(qBs/m). \quad (12.3)$$

Integrating (12.3) for a complex coordinate $Z(s)$ defined by $dZ/ds = P/m$ and $Z(0) = 0$ gives

$$Z(s) = \frac{m}{qB} \sin(qBs/m) + i \frac{m}{qB} [1 - \cos(qBs/m)] \sigma_3. \quad (12.4)$$

The real energy in (12.3) changes sign, and the time in (12.4) ceases to increase, despite $\det P = m^2$. This fails the ordinary at-rest electric charge limit even with zero magnetic charge. Determinant conservation alone is insufficient for a physical mass shell or causality.

12.2 Testing complex spacetime and its null set

Let $Z = X + iY$ be a complex coordinate, with real paravectors $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ and $Y = u + \mathbf{w} \cdot \boldsymbol{\sigma}$. Here $t, u \in \mathbb{R}$ and $\mathbf{r}, \mathbf{w} \in \mathbb{R}^3$ are independent coordinates. Multiplication gives

$$\det Z = t^2 - \mathbf{r}^2 - u^2 + \mathbf{w}^2 + 2i(tu - \mathbf{r} \cdot \mathbf{w}). \quad (12.5)$$

Complex nullity in (12.5) requires both

$$\det X = \det Y, \quad tu = \mathbf{r} \cdot \mathbf{w}.$$

For example, $Z = \sigma_1 + i\sigma_2$ is complex-null but its real part is spacelike. Identifying every complex-null separation with observable light propagation would therefore permit real spacelike connections. Taking only the real determinant gives instead

$$\text{re}(\det(dZ)) = dt^2 - d\mathbf{r}^2 - du^2 + d\mathbf{w}^2, \quad (12.6)$$

The metric (12.6) has signature (4, 4). Either proposal needs an operationally defined real causal slice. Complex field values on real spacetime require no such change of coordinates.

12.3 Separate masses and small couplings

Consider a real four-vector field $A(t, \mathbf{r}) = a(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$. Assign proposed real masses $m_0, m_1 \geq 0$ to its scalar and third vector components:

$$\begin{aligned} (\square + m_0^2)a &= 0, \\ (\square + m_1^2)A_3 &= 0. \end{aligned}$$

For $L = \exp(\theta\sigma_3/2)$, define $\beta = \tanh \theta$ and $\gamma = \cosh \theta$ with real rapidity θ . The passive boost gives

$$\begin{aligned} a' &= \gamma(a - \beta A_3), \\ (\square' + m_0^2)a' &= -\gamma\beta(m_0^2 - m_1^2)A_3. \end{aligned} \quad (12.7)$$

Covariance of (12.7) for arbitrary fields forces $m_0^2 = m_1^2$, and similarly for every spatial component. Distinct masses can label distinct fields, but cannot label time and space entries of one Lorentz vector.

As a separate toy model, mix two scalar fields. Let $z = a + ib$, with $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$, and let $m > 0$ and h be real constants; h has units of mass squared. The real Lagrangian density

$$L = \frac{1}{2} \det(\partial a) + \frac{1}{2} \det(\partial b) - \frac{1}{2} m^2 (a^2 + b^2) - hab \quad (12.8)$$

has positive time-derivative terms, since $\det(\partial a) = (\partial_t a)^2 - (\partial_{\mathbf{r}} a)^2$. The equations of (12.8) and their diagonalizing real combinations are

$$\begin{aligned} (\square + m^2)a + hb &= 0, \\ (\square + m^2)b + ha &= 0, \\ a' &= (a + b)/\sqrt{2}, \\ b' &= (a - b)/\sqrt{2}. \end{aligned}$$

$$(\square + m^2 + h)a' = 0, \quad (12.9)$$

$$(\square + m^2 - h)b' = 0. \quad (12.10)$$

Equations (12.9) and (12.10) have squared masses $m^2 \pm h$, both nonnegative when $|h| \leq m^2$. The mixing term breaks continuous $z \mapsto e^{ic}z$ symmetry ($c \in \mathbb{R}$ constant), but leaves the principal operator $\square$ and its light cone unchanged. Small coupling does not create complex coordinates. The electroweak interaction likewise requires additional internal gauge and chiral structure.^[9]

By contrast, the alternative kinetic term

$$\frac{1}{2} \operatorname{re}(\det(\partial z)) = \frac{1}{2} \det(\partial a) - \frac{1}{2} \det(\partial b)$$

has real kinetic Hamiltonian density H :

$$H = \frac{1}{2}[(\partial_t a)^2 + (\partial_r a)^2] - \frac{1}{2}[(\partial_t b)^2 + (\partial_r b)^2]. \quad (12.11)$$

The density (12.11) is unbounded below: the independent b mode is a ghost, whose negative kinetic sign persists at arbitrarily small coupling. ^[10]

12.4 What would be needed for a dark sector?

A dark-matter model must supply abundance, lifetime, gravitational clustering, and interaction rates. Complex notation alone fixes none. To distinguish matter from accelerated expansion, additionally assume homogeneous general-relativistic cosmology. Let $\phi(t) \in \mathbb{R}$ be a canonical scalar field, $V(\phi)$ its real potential-energy density, $u(t)$ its energy density, and $p(t)$ its isotropic pressure. Its stress tensor gives^[11]

$$u = \frac{1}{2}(\partial_t \phi)^2 + V, \quad (12.12)$$

$$p = \frac{1}{2}(\partial_t \phi)^2 - V. \quad (12.13)$$

Equations (12.12) and (12.13) are the established stress-tensor relations used to test the model. For real mass $m > 0$, take $V = m^2 \phi^2/2$ and oscillations faster than cosmic expansion: $\phi \simeq A \cos(mt)$ over a cycle, with slowly varying real amplitude A . Cycle averaging gives

$$\begin{aligned} \langle \frac{1}{2}(\partial_t \phi)^2 \rangle &= \langle V \rangle = m^2 A^2/4, \\ \langle p \rangle &= 0. \end{aligned}$$

This behaves as pressureless matter. Potential domination instead gives $p \simeq -u$. For the real positive cosmic scale factor $a(t)$ and Newton's constant $G > 0$, the acceleration equation is

$$\frac{\partial_t^2 a}{a} = -\frac{4\pi G}{3}(u + 3p). \quad (12.14)$$

Equation (12.14) requires $u + 3p < 0$ for acceleration, not merely an additional positive mass. A complex-spacetime hypothesis still needs a stable action, a physical causal structure, and distinct testable predictions.

A Pauli-matrix representation

This appendix translates the established paravector operations into matrices. Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$, and write $\mathbf{M} = \operatorname{mat}(Z) \in \mathbb{C}^{2 \times 2}$. The identity matrices are $\mathbf{I}_2$ and $\mathbf{I}_4$ in their stated dimensions.

A.1 Faithful representation

The Pauli images of the generators are

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Consequently,

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{Bmatrix}. \quad (\text{A.1})$$

The images in (A.1) obey the sigma product rule and are linearly independent over $\mathbb{C}$ together with $\mathbf{I}_2$. The map is therefore faithful, preserves addition, sends 1 to $\mathbf{I}_2$, and satisfies

$$\text{mat}(Z'Z) = \text{mat}(Z') \text{mat}(Z). \quad (\text{A.2})$$

Here Z' is any second complex paravector; (A.2) identifies algebra multiplication with ordinary matrix multiplication.

A.2 Conjugations and product order

For complex matrices $\mathbf{M}, \mathbf{N}$, transpose T exchanges row and column indices; entrywise conjugation C sends each $a + ib$ to $a - ib$ for real a, b . Their composition is $\mathbf{M}^H = (\mathbf{M}^C)^T$. Introduce the fixed real matrix

$$\mathbf{S} = \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}, \quad \mathbf{S}^{-1} = -\mathbf{S}.$$

The paravector sigma conjugation has the matrix image

$$\text{mat}(Z^*) = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S} =: \mathbf{M}^*. \quad (\text{A.3})$$

Substituting the basis in (A.3) gives

Algebra basis	1	i	σ_k	$i\sigma_k$
Hermitian image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$\text{mat}(\sigma_k)$	$-i \text{mat}(\sigma_k)$
Sigma-conjugate image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$-\text{mat}(\sigma_k)$	$i \text{mat}(\sigma_k)$

Here $k \in \{1, 2, 3\}$. These are exactly the operations of Section 2, so $\text{mat}(Z^H) = \mathbf{M}^H$. Their product-order rules are

$$\begin{aligned} (\mathbf{MN})^T &= \mathbf{N}^T \mathbf{M}^T, \\ (\mathbf{MN})^C &= \mathbf{M}^C \mathbf{N}^C, \\ (\mathbf{MN})^H &= \mathbf{N}^H \mathbf{M}^H. \end{aligned}$$

For sigma conjugation the middle factors cancel:

$$(\mathbf{MN})^* = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S} \mathbf{S}^{-1} \mathbf{N}^C \mathbf{S}.$$

Thus sigma conjugation preserves product order:

$$(\mathbf{MN})^* = \mathbf{M}^* \mathbf{N}^*. \quad (\text{A.4})$$

Equation (A.4) verifies (2.2) in matrices. Both H and * are commuting involutions. The matrix-only operations can be recovered as

$$\begin{aligned} \mathbf{M}^C &= \mathbf{S} \mathbf{M}^* \mathbf{S}^{-1}, \\ \mathbf{M}^T &= \mathbf{S} (\mathbf{M}^*)^H \mathbf{S}^{-1}. \end{aligned}$$

A.3 Scalar part, determinant, and norm

For $\mathbf{M} = (m_{jk})$, its trace is $m_{11} + m_{22}$ and its determinant is $m_{11}m_{22} - m_{12}m_{21}$. Substituting (A.1) gives

$$\text{sc}(Z) = \frac{1}{2} \text{tr} \mathbf{M} = S, \quad (\text{A.5})$$

$$\begin{aligned} (\text{vec}(Z))_k &= \frac{1}{2} \text{tr}(\text{mat}(\sigma_k) \mathbf{M}) = V_k, \\ \det Z &= \det \mathbf{M} = S^2 - \mathbf{V}^2. \end{aligned} \quad (\text{A.6})$$

By (A.5), matrix-trace cyclicity is precisely the scalar-part cyclicity used in the main text. For $\mathbf{M}' = \text{mat}(Z')$,

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}' \mathbf{M}) = \text{sc}(ZZ').$$

The remaining extraction and norm formulas are

$$\begin{aligned} \text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) &= \mathbf{M} - \text{sc}(Z) \mathbf{I}_2, \\ \text{mat}(\text{re}(Z)) &= \frac{1}{2}(\mathbf{M} + \mathbf{M}^H), \end{aligned} \quad (\text{A.7})$$

$$\text{mat}(\text{im}(Z)) = (\mathbf{M} - \mathbf{M}^H)/(2i), \quad (\text{A.8})$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(\mathbf{M}^H \mathbf{M}) = \|S\|^2 + \|\mathbf{V}\|^2. \quad (\text{A.9})$$

Equations (A.7)–(A.8) extract Hermitian matrices, so a real paravector has a Hermitian image. The norm in (A.9) is half the squared Frobenius norm; the determinant in (A.6) may be complex.

The matrix adjugate is

$$\text{adj}(\mathbf{M}) = (\text{tr} \mathbf{M}) \mathbf{I}_2 - \mathbf{M} = \text{mat}(\text{adj}(Z)) = (\mathbf{M}^*)^H. \quad (\text{A.10})$$

Equation (A.10) satisfies

$$\text{mat}(\text{adj}(Z) Z) = \det(\mathbf{M}) \mathbf{I}_2.$$

Consequently $\mathbf{M}^{-1} = \text{adj}(\mathbf{M}) / \det \mathbf{M}$ when the determinant is nonzero. The determinant product and adjugate order rules are the same in both representations. The quadratic identity (3.25) is precisely Cayley–Hamilton for a 2×2 matrix.

A.4 Matrix amplitudes and column spinors

Let ψ, ψ' be complex paravector amplitudes, with matrix images $\mathbf{M}_\psi, \mathbf{M}_{\psi'}$, and let $Z = Z^H$ be a real observable. For fields these depend on real $(t, \mathbf{r})$. The inner products (8.1) and (8.2) become

$$\begin{aligned} \langle \psi' | \psi \rangle &= \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_\psi), \\ \langle \psi' | Z | \psi \rangle &= \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_\psi). \end{aligned}$$

An unrestricted $\mathbf{M}_\psi$ has two independently transforming columns. The fixed ideal $\psi\Pi_3 = \psi$, with $\Pi_3 = (1 + \sigma_3)/2$, sets the second column to zero because $\text{mat}(\Pi_3) = \text{diag}(1, 0)$. Its conventional Pauli column is

$$\psi := \frac{1}{\sqrt{2}} \mathbf{M}_\psi \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}. \quad (\text{A.11})$$

Applying (A.11) to the normalized basis ψ_+, ψ_- of (8.15), the complete correspondence is

$$\psi = c_+ \psi_+ + c_- \psi_- \quad \longleftrightarrow \quad \mathbf{M}_\psi = \sqrt{2} \begin{Bmatrix} c_+ & 0 \\ c_- & 0 \end{Bmatrix} \quad \longleftrightarrow \quad \psi = \begin{Bmatrix} c_+ \\ c_- \end{Bmatrix}.$$

Here $c_+, c_- \in \mathbb{C}$, or complex scalar functions of $(t, \mathbf{r})$. Substitution cancels the factor of two from the trace pairing:

$$\begin{aligned} \langle \psi' | \psi \rangle &= \psi'^H \psi, \\ \langle \psi' | Z | \psi \rangle &= \psi'^H \text{mat}(Z) \psi. \end{aligned}$$

Left action $T\psi$ maps to $\text{mat}(T)\psi$ for any complex paravector T . For nonzero norm $\rho = \psi^H \psi$, the real state paravector B and projectors $\Pi_\pm = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ ($\mathbf{u} \in \mathbb{R}^3$ a unit measurement axis) translate to

$$\frac{1}{2} \text{mat}(B) = \psi \psi^H / \rho, \quad (\text{A.12})$$

$$p_\pm(\mathbf{u}) = \psi^H \text{mat}(\Pi_\pm) \psi / \rho. \quad (\text{A.13})$$

Equations (A.12) and (A.13) are the ordinary density matrix and Born rule. An unrestricted matrix amplitude instead gives $\mathbf{M}_\psi \mathbf{M}_\psi^H / \text{tr}(\mathbf{M}_\psi \mathbf{M}_\psi^H)$, which is pure exactly when $\mathbf{M}_\psi$ has rank one.

A.5 Dirac columns and currents

For a Dirac block $\Psi = \{\Psi_1, \Psi_2\}^T$, let $\psi_1, \psi_2 \in \mathbb{C}^2$ be the corresponding ideal columns. Then

$$\mathbf{W}(Z)\Psi \quad \longleftrightarrow \quad \begin{Bmatrix} 0 & \text{mat}(Z) \\ \text{mat}(\text{adj}(Z)) & 0 \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \end{Bmatrix}.$$

The usual Dirac-adjoint row is $\{\psi_2^H, \psi_1^H\}$ in this chiral ordering. The real current and rotation-spin state become

$$\rho = \psi_1^H \psi_1 + \psi_2^H \psi_2, \quad (\text{A.14})$$

$$J_k = -\psi_1^H \text{mat}(\sigma_k) \psi_1 + \psi_2^H \text{mat}(\sigma_k) \psi_2, \quad (\text{A.15})$$

$$\frac{1}{2} \text{mat}(B) = \frac{\psi_1 \psi_1^H + \psi_2 \psi_2^H}{\rho}. \quad (\text{A.16})$$

Equations (A.14)–(A.15) are the real components of $J = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$. The spin state (A.16) requires $\rho > 0$. For rest-spin statistics, apply the matrix image of $\mathbf{T}^H$ from (8.85) before forming the same sum.

Recall $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real angle or rapidity θ . The chiral transformation laws are

$$\begin{Bmatrix} \psi'_1 \\ \psi'_2 \end{Bmatrix} = \begin{Bmatrix} \text{mat}(R)^H & 0 \\ 0 & \text{mat}(R)^H \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \end{Bmatrix}, \quad (\text{A.17})$$

$$\begin{Bmatrix} \psi'_1 \\ \psi'_2 \end{Bmatrix} = \begin{Bmatrix} \text{mat}(L) & 0 \\ 0 & \text{mat}(L)^{-1} \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \end{Bmatrix}. \quad (\text{A.18})$$

Equations (A.17)–(A.18) give equal rotation generators and opposite boost generators, as in (9.4)–(9.5).

Finally, the ideal map sends $\psi^*\sigma_1$ to $\mathbf{S}^{-1}\psi^{\mathbf{C}}$, since $\mathbf{S} = \text{mat}(\text{i}\sigma_2)$. The charge-conjugate block of (8.102) therefore is

$$\Psi' \longleftrightarrow \begin{Bmatrix} \mathbf{S}^{-1}\psi_2^{\mathbf{C}} \\ -\mathbf{S}^{-1}\psi_1^{\mathbf{C}} \end{Bmatrix}. \quad (\text{A.19})$$

The prime in (A.19) denotes charge reversal.

A.6 Four components and the standard Dirac basis

Write $\Psi_1 = a\psi_+ + b\psi_-$ and $\Psi_2 = c\psi_+ + d\psi_-$, where $a, b, c, d : \mathbb{R}^4 \rightarrow \mathbb{C}$ are the four scalar components in the normalized ideal basis. Recall real mass $m > 0$, real charge q , and real potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. The mechanical energy and momentum operators are $E = \text{i}\partial_t - qV$ and $\mathbf{p} = -\text{i}\partial_{\mathbf{r}} - q\mathbf{A}$. The coupled paravector equations give exactly

$$(E - p_3)c - (p_1 - \text{i}p_2)d = ma, \quad (\text{A.20})$$

$$-(p_1 + \text{i}p_2)c + (E + p_3)d = mb, \quad (\text{A.21})$$

$$(E + p_3)a + (p_1 - \text{i}p_2)b = mc, \quad (\text{A.22})$$

$$(p_1 + \text{i}p_2)a + (E - p_3)b = md. \quad (\text{A.23})$$

Operators in (A.20)–(A.23) act on the scalar immediately to their right. Their Hamiltonian on $\{a, b, c, d\}^{\mathbf{T}}$ is

$$H = \begin{Bmatrix} qV - p_3 & -p_1 + \text{i}p_2 & m & 0 \\ -p_1 - \text{i}p_2 & qV + p_3 & 0 & m \\ m & 0 & qV + p_3 & p_1 - \text{i}p_2 \\ 0 & m & p_1 + \text{i}p_2 & qV - p_3 \end{Bmatrix}. \quad (\text{A.24})$$

This is the component image of (8.77).

To identify conventional gamma matrices, put $\sigma_k = \text{mat}(\sigma_k)$ for $k = 1, 2, 3$ and define

$$\gamma^0 = \begin{Bmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{Bmatrix}, \quad (\text{A.25})$$

$$\gamma^k = \begin{Bmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{Bmatrix}. \quad (\text{A.26})$$

Equations (A.25)–(A.26) obey $(\gamma^0)^2 = \mathbf{I}_4$, $(\gamma^k)^2 = -\mathbf{I}_4$, and $\gamma^j\gamma^k + \gamma^k\gamma^j = 0$ for distinct $j, k \in \{0, 1, 2, 3\}$. Thus

$$\mathbf{W}(\text{i}\partial - q\Phi^*) \longleftrightarrow \gamma^0 E - \sum_{k=1}^3 \gamma^k p_k, \quad (\text{A.27})$$

so (A.27) is the standard chiral Dirac operator with signature $(+, -, -, -)$.^[4] Here $\Phi = V + \mathbf{A} \cdot \sigma$ is the real potential paravector.

The unitary change to the standard Dirac basis, followed by removal of the rest phase, defines the envelope columns

$$\begin{Bmatrix} \phi \\ \chi \end{Bmatrix} = \frac{e^{\text{i}mt}}{\sqrt{2}} \begin{Bmatrix} \mathbf{I}_2 & \mathbf{I}_2 \\ -\mathbf{I}_2 & \mathbf{I}_2 \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \end{Bmatrix}. \quad (\text{A.28})$$

Here $\phi, \chi \in \mathbb{C}^2$ are the column images of the ideal fields $\phi = e^{imt}(\Psi_1 + \Psi_2)/\sqrt{2}$ and $\chi = e^{imt}(\Psi_2 - \Psi_1)/\sqrt{2}$. Before removing the rest phase, the basis change in (A.28) sends (A.24) to

$$H' = \left\{ \begin{array}{cc} (qV + m)\mathbf{I}_2 & \sum_k p_k \boldsymbol{\sigma}_k \\ \sum_k p_k \boldsymbol{\sigma}_k & (qV - m)\mathbf{I}_2 \end{array} \right\}. \quad (\text{A.29})$$

Removing the rest energy from (A.29) gives the exact envelope equations

$$\begin{aligned} E\phi &= \sum_k p_k \boldsymbol{\sigma}_k \chi, \\ (E + 2m)\chi &= \sum_k p_k \boldsymbol{\sigma}_k \phi. \end{aligned}$$

These identify the large and small amplitudes used in the Pauli limit.

For real electric and magnetic components E_k, B_k , the squared Dirac field coupling has the explicit image

$$\begin{aligned} \text{mat}(iqF) &= q \left\{ \begin{array}{cc} iE_3 - B_3 & iE_1 + E_2 - B_1 + iB_2 \\ iE_1 - E_2 - B_1 - iB_2 & -iE_3 + B_3 \end{array} \right\}, \\ \text{mat}(iqF^*) &= \text{mat}(iqF)^H. \end{aligned} \quad (\text{A.30})$$

The indexed E_k in (A.30) are field components; unindexed E remains the mechanical energy operator. The two couplings are $-q\mathbf{B} \cdot \boldsymbol{\sigma} \pm iq\mathbf{E} \cdot \boldsymbol{\sigma}$.

A.7 Quaternion subalgebra

Define $q_k = -i\sigma_k$ for $k = 1, 2, 3$, and $\mathbf{q} = \{q_1, q_2, q_3\}$. These are bivectors, not electric charges. Their product follows from the sigma rule:

$$q_j q_k = -\delta_{jk} + \sum_{l=1}^3 \epsilon_{jkl} q_l. \quad (\text{A.31})$$

Equation (A.31) gives $q_k^2 = -1$, $q_1 q_2 = q_3$, and $q_1 q_2 q_3 = -1$. For real a and $\mathbf{A} \in \mathbb{R}^3$, the quaternion

$$Q = a + \mathbf{A} \cdot \mathbf{q} = a - i\mathbf{A} \cdot \boldsymbol{\sigma} \quad (\text{A.32})$$

is a complex paravector with real quaternion coordinates. Elements (A.32) form the real scalar-bivector subalgebra. For a second such element $Q' = a' + \mathbf{A}' \cdot \mathbf{q}$, expansion gives

$$QQ' = (aa' - \mathbf{A} \cdot \mathbf{A}') + (a\mathbf{A}' + a'\mathbf{A} + \mathbf{A} \times \mathbf{A}') \cdot \mathbf{q}.$$

Because $q_k^H = -q_k$ and $q_k^* = q_k$, quaternion conjugation is the existing Hermitian operation:

$$\begin{aligned} Q^H &= a - \mathbf{A} \cdot \mathbf{q}, \\ Q^* &= Q, \\ \text{adj}(Q) &= Q^H, \\ QQ^H &= Q^H Q = a^2 + \mathbf{A}^2 = \det Q. \end{aligned} \quad (\text{A.33})$$

For nonzero Q , (A.33) gives the inverse $Q^{\mathbf{H}}/(a^2 + \mathbf{A}^2)$. The explicit matrix image is

$$\text{mat}(Q) = \begin{Bmatrix} a - \mathbf{i}A_3 & -A_2 - \mathbf{i}A_1 \\ A_2 - \mathbf{i}A_1 & a + \mathbf{i}A_3 \end{Bmatrix}. \quad (\text{A.34})$$

In (A.34), $\text{mat}(q_k) = -\mathbf{i} \text{mat}(\sigma_k)$. The conjugation and norm identities become

$$\begin{aligned} \text{mat}(Q^{\mathbf{H}}) &= \text{mat}(Q)^{\mathbf{H}}, \\ \text{mat}(Q)^{\mathbf{H}} \text{mat}(Q) &= (a^2 + \mathbf{A}^2) \mathbf{I}_2, \\ \det(\text{mat}(Q)) &= a^2 + \mathbf{A}^2. \end{aligned}$$

A unit quaternion therefore corresponds to a unitary matrix of determinant one. For real unit $\mathbf{u}$ and angle θ , let

$$Q = \cos(\theta/2) + \mathbf{u} \cdot \mathbf{q} \sin(\theta/2).$$

Then $Q = R^{\mathbf{H}}$ for $R = \exp(\mathbf{i}\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$. For a real vector $\mathbf{r}$, the quaternion rotation

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^{\mathbf{H}} \quad (\text{A.35})$$

identifies (A.35) with Rodrigues' formula (4.11).

B Notation and conventions

B.1 Reading conventions

- Natural units are $\hbar = c = 1$, with metric signature $(+, -, -, -)$. The central pseudoscalar $\mathbf{i} = \sigma_1 \sigma_2 \sigma_3$ maps to the ordinary matrix complex unit; unindexed e denotes Euler's number.
- Vectors and matrices are bold; real paravectors have real scalar and vector coefficients. Curly braces display component lists and matrices. A vector square means its dot product with itself.
- $\text{sc}(Z)$ is the scalar coefficient and $\text{vec}(Z)$ its vector coefficients. $\text{re}(Z)$ and $\text{im}(Z)$ are real paravectors. The positive norm $\|Z\|^2$ is distinct from the possibly complex determinant $\det Z$.
- $^{\mathbf{H}}$ reverses product order, $*$ preserves it, and $\text{adj}(Z) = Z^{*\mathbf{H}}$. The matrix-only $^{\mathbf{T}}$ and $^{\mathbf{C}}$ mean transpose and entrywise conjugation.
- A unit has a two-sided inverse. Proper velocity is a real determinant-one unit: $U^{\mathbf{H}} = U$ and $U^{-1} = U^*$. Hermitian unitarity instead means $U^{\mathbf{H}}U = 1$.
- Differential operators act to the right within their displayed grouping. Gauge-covariant energy and momentum do not commute, so their ordered products retain electromagnetic field terms. Hermiticity of differential operators additionally requires a domain, boundary conditions, and integration by parts.
- On Dirac blocks, $^{\mathbf{H}}$ transposes the block and adjoints each entry; $*$ acts entrywise. The fixed right projector Π_3 selects one ideal. Left projectors $\Pi_{\pm}$ select spin outcomes; these are distinct actions even when their algebra elements coincide.
- Primes denote locally specified transformations; q' is the explicitly defined magnetic-charge exception. Plus/minus and parallel/perpendicular subscripts label physical alternatives and projections. Scalar Lagrangians and Hamiltonians are defined locally as L and H .

B.2 Changes from the original notes

The algebra and its operations are retained. The following changes separate mathematical types and remove collisions; all symbols are defined locally at first use.

Original notation	Present notation	Reason
Reversal Z^R	Z^H	Matches matrix Hermitian conjugation; frees R for the rotor.
Double-angle quadratic form	$\det Z$	Distinguishes the Minkowski form from the positive norm.
scalar, vector	sc, vec	Short names for the same coefficient extractions.
real, img	re, im	Short names for the same real paravector parts.
Spatial momentum P	$\boldsymbol{p}$	Reserves P for the full four-momentum paravector.
Electric/magnetic script letters	$\boldsymbol{E}, \boldsymbol{B}$	Ordinary bold vectors; scalar E is energy, including rest energy.
Wave amplitude Ψ in several roles	$\psi, \boldsymbol{\Psi}, \phi, \chi$	Distinguishes scalar/ideal amplitudes, a two-entry Dirac block, and its large/small envelopes.
KG frequency operator Ω	H	Identifies the positive free Hamiltonian; Ω is the local frame generator.
Differential blocks D, D'	$\boldsymbol{D}, \boldsymbol{D}'$	Bold type marks the two-entry operators; D remains the electromagnetic source paravector.
Frequency and spin branch labels	$\psi_{\pm}$ and ψ_+, ψ_-	KG uses scalar frequency fields; spin uses a fixed ideal basis. The containing subsection fixes the type.
Common transform T	R, L , or general T	Names rotation and boost factors and keeps their actions distinct.

In the added quantum examples, ρ is density, ω canonical mode frequency, and N oscillator level. The product $\boldsymbol{p} \cdot \boldsymbol{\sigma}$ is written directly; K denotes energy above rest. Magnetic charge is q' . These choices avoid using unindexed e , density n , current j , or K for an operator that is not kinetic energy.

Two changes correct claims rather than notation. Proper velocity obeys (4.5), whereas Hermitian unitarity means (3.14). A real spacetime boost uses the congruence (4.12); determinant-preserving similarity alone does not give that action. The quantum derivations also retain the block transpose in Hermitian conjugation and distinguish signed KG charge density from positive Dirac probability density.

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C Equation reference

Each displayed formula repeats a numbered result; its right-hand number links to the definition or derivation. Natural units are $\hbar = c = 1$. The linked text supplies domains, component expansions, and approximation conditions. Speculative models are confined to Section 12.

Algebra. Recall complex paravectors $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, with complex scalar S, S' and complex vectors $\mathbf{V}, \mathbf{V}'$. The prime here labels a second element.

$$\begin{aligned} ZZ' &= (SS' + \mathbf{V} \cdot \mathbf{V}') \\ &\quad + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \end{aligned} \tag{3.28}$$

$$Z^H = S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} \tag{3.5}$$

$$Z^* = S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} \tag{3.4}$$

$$\text{adj}(Z) = Z^{*H} = S - \mathbf{V} \cdot \boldsymbol{\sigma} \tag{3.6}$$

$$\text{tr } Z = 2 \text{sc}(Z) \tag{3.7}$$

$$\text{re}(Z) = \frac{1}{2}(Z + Z^H) \tag{3.19}$$

$$\text{im}(Z) = (Z - Z^H)/(2i) \tag{3.20}$$

$$\det Z = \text{adj}(Z) Z = S^2 - \mathbf{V}^2 \tag{3.8}$$

$$Z^{-1} = \text{adj}(Z) / \det Z \quad (\det Z \neq 0) \tag{3.12}$$

$$\|Z\|^2 = \text{sc}(Z^H Z). \tag{3.34}$$

The coefficient form is (3.33); product-order and cyclicity rules are (2.1), (2.2), and (3.18). Projectors and the spectral rule are (3.36) and (3.40). Matrix and quaternion images are (A.1) and (A.34).

Rotations and boosts. Here $\mathbf{u}$ is a real unit axis and θ a real angle or rapidity; $\beta = \tanh \theta$ and $\gamma = \cosh \theta$ in the boost formulas.

$$R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cos(\theta/2) + i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma} \tag{3.50}$$

$$L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cosh(\theta/2) + \sinh(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma} \tag{3.51}$$

$$L = \sqrt{\frac{\gamma+1}{2}} \left(1 + \frac{\gamma\beta}{\gamma+1} \mathbf{u} \cdot \boldsymbol{\sigma} \right) \tag{3.66}$$

$$L^2 = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}) \tag{3.67}$$

$$Z' = R^H Z R \tag{4.10}$$

$$Z' = L^{-1} Z L^{-1}. \tag{4.12}$$

The last line is the passive boost of a real four-vector. Rodrigues' rotation is (4.11); boost components are (4.13)–(4.14). The general determinant-one congruence is (4.8); physical field and spinor laws are in Section 9.

Spacetime and particles. Recall real event X , proper time s , coordinate velocity $\mathbf{v}$, and constant mass $m > 0$. The real four-momentum P has mechanical energy E (including rest energy) and spatial momentum $\mathbf{p}$.

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \quad (5.1)$$

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2 \quad (5.3)$$

$$ds^2 = \det(dX) = dt^2 - d\mathbf{r}^2 \quad (6.1)$$

$$U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) \quad (6.4)$$

$$U^* U = 1 \quad (4.5)$$

$$P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma} \quad (6.6)$$

$$E^2 - \mathbf{p}^2 = m^2. \quad (6.7)$$

Here $\gamma = dt/ds = (1 - \mathbf{v}^2)^{-1/2}$. Proper velocity is real, with unit determinant; Hermitian unitarity is the separate condition (3.14). Elapsed time, acceleration, and particle action are (6.5), (6.8), and (6.13).

Electromagnetism. Recall real potentials $V, \mathbf{A}$, real electric and magnetic fields $\mathbf{E}, \mathbf{B}$, the complex field F , real potential paravector Φ , and real charge-current paravector D . The real gauge scalar is S ; q is charge. These fields are functions of real spacetime.

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad (7.1)$$

$$\partial F = D^* \quad (7.2)$$

$$\square F = \partial^* D^* \quad (7.7)$$

$$\partial \Phi = S + F^* \quad (7.11)$$

$$F^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B} \quad (7.19)$$

$$\frac{dP}{ds} = q \operatorname{re}(FU) \quad (7.26)$$

$$T(N) = \frac{1}{2} F N F^H \quad (7.30)$$

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.32)$$

Here N is a real paravector argument of the stress map, u field energy density, $\mathbf{S}$ energy flux, and $\mathbf{J}$ electric current density. Potential components are (7.12)–(7.13); force components are (7.22) and (7.24). The action derivation uses (7.17) and (7.20).

Spin and probabilities. Recall complex ideal amplitudes ψ, ψ' in the fixed right ideal, with $\psi \neq 0$, real norm density $\rho > 0$, real state paravector B , and real unit measurement axis $\mathbf{u}$. The real observable Z acts from the left.

$$\langle \psi' | \psi \rangle = \operatorname{sc}(\psi'^H \psi) \quad (8.1)$$

$$\rho = \operatorname{sc}(\psi^H \psi) \quad (8.3)$$

$$B = \psi \psi^H / \rho = 1 + \mathbf{s} \cdot \boldsymbol{\sigma} \quad (8.18)$$

$$p_{\pm}(\mathbf{u}) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}) \quad (8.20)$$

$$\langle Z \rangle_{\psi} = \operatorname{sc}(\psi^H Z \psi) / \rho. \quad (8.4)$$

Here $\mathbf{s}$ is the real Bloch vector, of length one for a pure state. The normalized ideal basis is (8.15); spin zero, one-half, and one are derived in Sections 8.1.3–8.1.5. The density matrix and Born probabilities translate exactly as (A.12) and (A.13). Spatial probabilities use the integrated pairing (8.21).

Relativistic waves. Recall real mass $m > 0$, charge q , and potentials $V, \mathbf{A}$. Their real potential paravector is $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. Here $E, \mathbf{p}$ denote mechanical differential operators, ψ a complex scalar KG field, and Ψ a two-entry complex ideal Dirac field.

$$E = i\partial_t - qV \quad (8.28)$$

$$\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A} \quad (8.29)$$

$$(\mathbf{p} \cdot \boldsymbol{\sigma})^2 = \mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma} \quad (8.33)$$

$$[E, \mathbf{p} \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma} \quad (8.34)$$

$$(-E^2 + \mathbf{p}^2 + m^2)\psi = 0 \quad (8.42)$$

$$\mathbf{W}(Z) = \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix} \quad (8.59)$$

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0 \quad (8.69)$$

$$J = \frac{1}{m} \text{re}(\psi^H(E + \mathbf{p} \cdot \boldsymbol{\sigma})\psi) \quad (8.46)$$

$$J = \text{adj}(\Psi_1 \Psi_1^H) + \Psi_2 \Psi_2^H. \quad (8.78)$$

Here Z is a complex paravector, $\mathbf{I}$ the block identity, and Ψ_1, Ψ_2 the ideal entries of Ψ . The last two lines define real KG and Dirac currents respectively. The KG density is signed; Dirac density is nonnegative and normalizes by (8.81). The free equations are (8.37) and (8.62); the two squared interacting entries are (8.106) and (8.107). Free Fourier solutions are (8.39) and (8.89); complete massive reconstructions are (8.67) and (8.110). The free unitary map (8.85) gives conserved rest spin.

Low-energy limits and examples. After removing the rest phase, the complex scalar envelope ψ obeys Schrödinger's equation to leading order:

$$i\partial_t \psi = \left(qV + \frac{\mathbf{p}^2}{2m} \right) \psi. \quad (8.55)$$

The complex ideal envelope ϕ instead obeys Pauli's equation:

$$i\partial_t \phi = \left(qV + \frac{\mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \right) \phi. \quad (8.118)$$

For static smooth fields with the stated small-velocity counting, the normalized positive-energy envelope has Hamiltonian

$$\begin{aligned} H = qV + \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2}{2m} - \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^4}{8m^3} \\ - \frac{q}{8m^2} \partial_{\mathbf{r}} \cdot \mathbf{E} - \frac{q}{8m^2} (\mathbf{E} \times \mathbf{p} - \mathbf{p} \times \mathbf{E}) \cdot \boldsymbol{\sigma} + O(mv^6). \end{aligned} \quad (8.126)$$

The small parameter v and retained operator order are specified there. Worked solutions are in Sections 8.5.1 (Stern–Gerlach and precession), 8.5.2 (constant potentials), and 8.5.3 (magnetic levels).

Magnetic duality and constant-field motion. For real magnetic-current paravector M , Maxwell's extension is

$$\partial F = D^* + \mathrm{i}M^*. \quad (11.4)$$

For real magnetic charge q' and real proper velocity U , the force is

$$\frac{\mathrm{d}P}{\mathrm{d}s} = \mathrm{re}((q - \mathrm{i}q')FU). \quad (11.1)$$

For constant F and real future initial momentum with mass $m > 0$, define

$$T(s) = \exp\left(\frac{(q - \mathrm{i}q')s}{2m}F\right). \quad (11.11)$$

The real future mass-shell solution is

$$P(s) = T(s)P(0)T(s)^{\mathrm{H}}. \quad (11.12)$$

This classical formalism retains real spacetime. The separate complex-coordinate hypotheses and their failed ordinary limits are tested in Section 12.